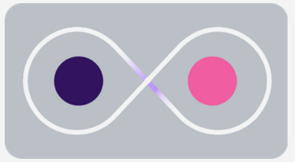
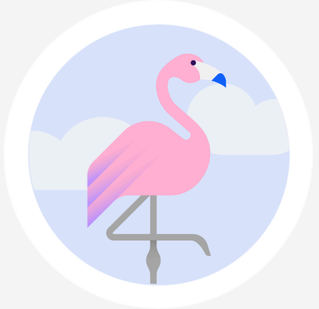
Qiskit Fall Fest 2026

Quantum Foundations







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IBM

Quantum Mechanics for Quantum Computing



Road Map for today!

What is a Qubit?	Dirac Notation	Bloch Sphere	Quantum Gates
Probability	Entanglement	Mathematical Framework	Circuits

print(“Hello Quantum World!”)

Quantum Quibits

- A Qubit, or a Quantum Qubit is the basic unit of information used to encode data in quantum computing.
- It is the quantum equivalent of the traditional bits used by classical computers to encode information in binary.
- Qubit in simple terms is a two state system. A familiar example being the spin of an electron, or Schrodingers cat being in a state of alive and dead.

Mere pass ~~Ma~~ Qubit hai!

Dirac (Bra-Ket) Notation



- In Quantum Mechanics and Computation, Bra-Ket notation is used to denote quantum states using angle brackets (<, >) and vertical bars (|).

- In quantum mechanics, a quantum state is typically represented as an element of a complex Hilbert space, for example, the infinite-dimensional vector space of all possible wavefunctions.

- It is common and useful in physics to denote an element ϕ of an abstract complex vector space as a ket $|\phi\rangle$, to refer to it as a "ket" rather than as a vector, and bra $\langle\phi|$ is an element of its dual space.

Qubits as Vectors

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

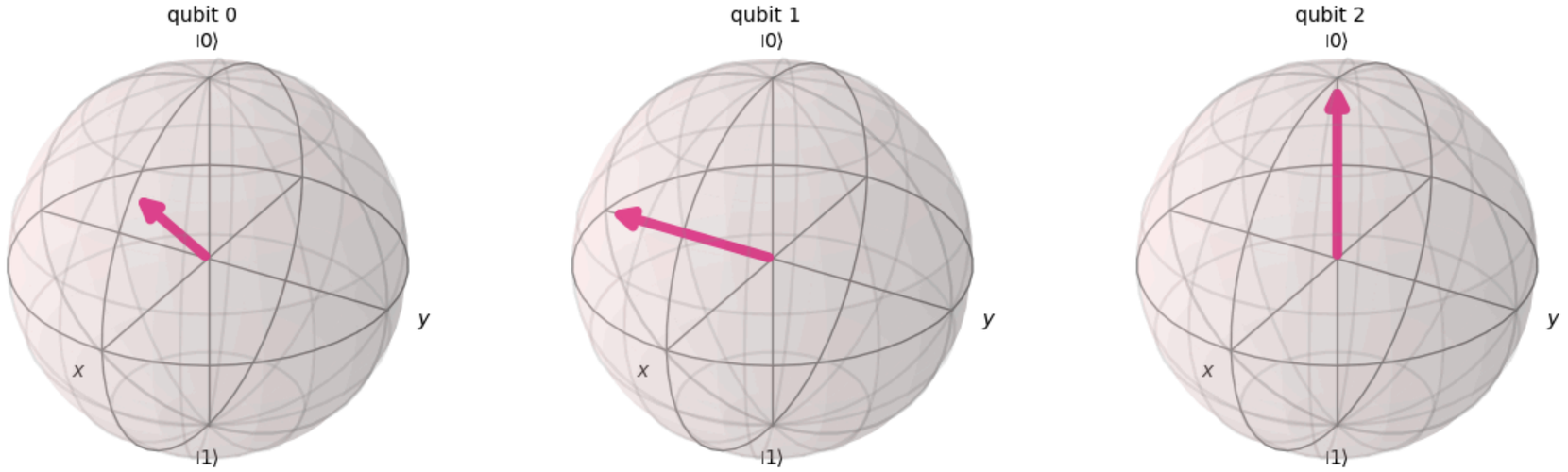
$$|00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, |01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix},$$

$$|10\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, |11\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Bloch Sphere

- Qubits are generally represented as two orthonormal vectors, and a state of a qubit is the superposition of these two.

$$\Psi = \cos(\frac{\theta}{2})|0\rangle + e^{i\phi}\sin(\frac{\theta}{2})|1\rangle$$



Quantum Gates

- The most commonly used gates for Quantum Teleportation are the three Pauli matrices X, Y, Z, the Hadamard gate, and the CNOT gate

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$X|0\rangle = |1\rangle, X|1\rangle = |0\rangle \quad Y|0\rangle = i|1\rangle, Y|1\rangle = -i|0\rangle \quad Z|0\rangle = |0\rangle, Z|1\rangle = -|1\rangle$$

Quantum Gates

- The most commonly used gates for Quantum Teleportation are the three Pauli matrices X, Y, Z, the Hadamard gate, and the CNOT gate

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$CNOT|00\rangle = |00\rangle, CNOT|01\rangle = |01\rangle$$
$$CNOT|10\rangle = |11\rangle, CNOT|11\rangle = |10\rangle$$

Measuring Probability

- The probability of the state is equal to the absolute value squared of the corresponding state vector entry.

- After measurement, the state of the system changes to the basis vector corresponding to the classical outcome.

- Example:

$$|\Psi\rangle = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = \sum_{i=1}^{n} \alpha_i |i\rangle \qquad \Pr(\text{outcome is } |i\rangle) = |\alpha_i|^2$$

Multiple Quibits and Entanglement

- Consider two quantum systems with Hilbert spaces H1 and H2, their combined Hilbert space is H1 ⊗ H2, and a state of this system is said to be entangled if it can not be represented as a dot product of the two individual Hilbert spaces.

$$|\psi\rangle_1 = |\uparrow\rangle|\downarrow\rangle \qquad |\psi\rangle_2 = |\uparrow\rangle|\downarrow\rangle + |\downarrow\rangle|\uparrow\rangle$$

Multiple Quibits and Entanglement

$$|\psi\rangle_1 = |\downarrow\rangle|\uparrow\rangle$$

$$|\psi\rangle_2 = |\uparrow\rangle|\downarrow\rangle + |\uparrow\rangle|\uparrow\rangle$$

$$|\psi\rangle_3 = |\downarrow\rangle|\downarrow\rangle + |\uparrow\rangle|\uparrow\rangle$$

$$|\psi\rangle_4 = |\uparrow\rangle|\downarrow\rangle - |\downarrow\rangle|\uparrow\rangle$$

Method behind the Madness

- Initially we begin with the assumption that both Alice's and Bob's is |0⟩. We can label them as |0⟩_B|0⟩_A.
- Hadamard on Alice → CNOT on Alice (control) and Bob = Bell States!

$$
\begin{aligned}
CNOT(A,B)H_A|0\rangle_B|0\rangle_A &= CNOT(A,B)|0\rangle_B\frac{1}{\sqrt{2}}(|0\rangle_A+|1\rangle_A) \\
&= \frac{1}{\sqrt{2}}(CNOT(A,B)|0\rangle_B|0\rangle_A + CNOT(A,B)|0\rangle_B|1\rangle_A) \\
&= \frac{1}{\sqrt{2}}(|0\rangle_B|0\rangle_A+|1\rangle_B|1\rangle_A)
\end{aligned}
$$

Method behind the Madness

- We now introduce the secret state prepared on Q:

$$|\psi\rangle_Q = \alpha_0|0\rangle_Q + \alpha_1|1\rangle_Q$$

- Keep in mind, at this point Q is simply next to Alice's qubit and no entanglement has taken place.

$$|\psi\rangle_{AB}|\psi\rangle_Q = \frac{1}{\sqrt{2}}(|0\rangle_B|0\rangle_A + |1\rangle_B|1\rangle_A)\,(\alpha_0|0\rangle_Q + \alpha_1|1\rangle_Q).$$

Method behind the Madness

- CNOT with Q (control) and Alice → Hadamard on Q

$$
\begin{aligned}
&H_Q\,\mathrm{CNOT}(Q,A)\,|\psi\rangle_{AB}|\psi\rangle_Q \\
&= H_Q\,\mathrm{CNOT}(Q,A)\,\frac{1}{\sqrt{2}}(|0\rangle_B|0\rangle_A + |1\rangle_B|1\rangle_A)\,(\alpha_0|0\rangle_Q + \alpha_1|1\rangle_Q) \\
&= H_Q\,\mathrm{CNOT}(Q,A)\,\frac{1}{\sqrt{2}}\Big(\alpha_0|0\rangle_B|0\rangle_A|0\rangle_Q + \alpha_1|0\rangle_B|0\rangle_A|1\rangle_Q + \alpha_0|1\rangle_B|1\rangle_A|0\rangle_Q + \alpha_1|1\rangle_B|1\rangle_A|1\rangle_Q\Big) \\
&= H_Q\,\frac{1}{\sqrt{2}}\Big(\alpha_0|0\rangle_B|0\rangle_A|0\rangle_Q + \alpha_1|0\rangle_B|1\rangle_A|1\rangle_Q + \alpha_0|1\rangle_B|1\rangle_A|0\rangle_Q + \alpha_1|1\rangle_B|0\rangle_A|1\rangle_Q\Big) \\
&= \frac{1}{2}\Big(\alpha_0|0\rangle_B|0\rangle_A|0\rangle_Q + \alpha_0|0\rangle_B|0\rangle_A|1\rangle_Q + \alpha_1|0\rangle_B|1\rangle_A|0\rangle_Q - \alpha_1|0\rangle_B|1\rangle_A|1\rangle_Q \\
&\qquad + \alpha_0|1\rangle_B|1\rangle_A|0\rangle_Q + \alpha_0|1\rangle_B|1\rangle_A|1\rangle_Q + \alpha_1|1\rangle_B|0\rangle_A|0\rangle_Q - \alpha_1|1\rangle_B|0\rangle_A|1\rangle_Q\Big)
\end{aligned}
$$

Method behind the Madness

- Let us group the terms above according to the outcomes of measurements on qubits A and Q:

$$
\begin{aligned}
|\psi\rangle = \frac{1}{2}\Big(& (\alpha_0|0\rangle_B + \alpha_1|1\rangle_B)|0\rangle_A|0\rangle_Q \\
& + (\alpha_0|0\rangle_B - \alpha_1|1\rangle_B)|0\rangle_A|1\rangle_Q \\
& + (\alpha_1|0\rangle_B + \alpha_0|1\rangle_B)|1\rangle_A|0\rangle_Q \\
& + (-\alpha_1|0\rangle_B + \alpha_0|1\rangle_B)|1\rangle_A|1\rangle_Q \Big)
\end{aligned}
$$

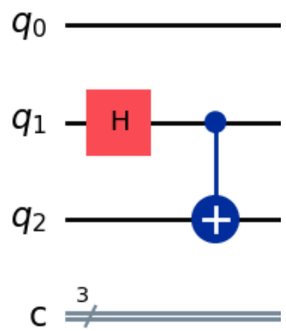
Method behind the Madness

- The table given below demonstrates how the outcome for Alice's qubits, collapse Bob's, and how we use gates X and Z to get back the state Q for Bob thus having transported the information.

Alice outcome	Bob's state	Instruction to Bob	Result
$\|0\rangle_A\|0\rangle_Q$	$\alpha_0\|0\rangle_B + \alpha_1\|1\rangle_B$	None	$\alpha_0\|0\rangle_B + \alpha_1\|1\rangle_B$
$\|0\rangle_A\|1\rangle_Q$	$\alpha_0\|0\rangle_B - \alpha_1\|1\rangle_B$	Z	$\alpha_0\|0\rangle_B + \alpha_1\|1\rangle_B$
$\|1\rangle_A\|0\rangle_Q$	$\alpha_1\|0\rangle_B + \alpha_0\|1\rangle_B$	X	$\alpha_0\|0\rangle_B + \alpha_1\|1\rangle_B$
$\|1\rangle_A\|1\rangle_Q$	$-\alpha_1\|0\rangle_B + \alpha_0\|1\rangle_B$	X then Z	$\alpha_0\|0\rangle_B + \alpha_1\|1\rangle_B$

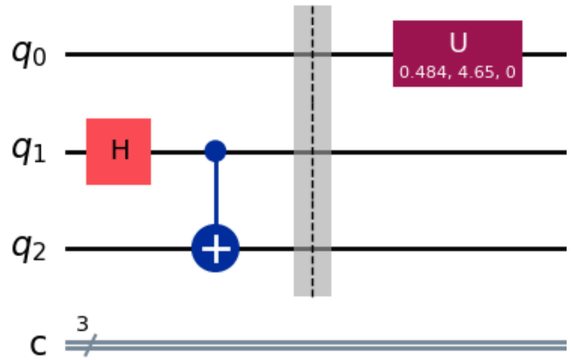
Circuit Diagrams

- Step 1: Entanglement of Alice and Bob’s qubit



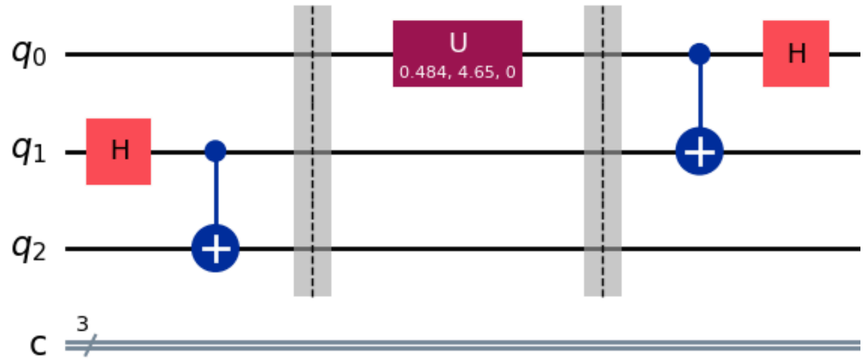
Circuit Diagrams

- Step 2: Generation fo secret state on Q



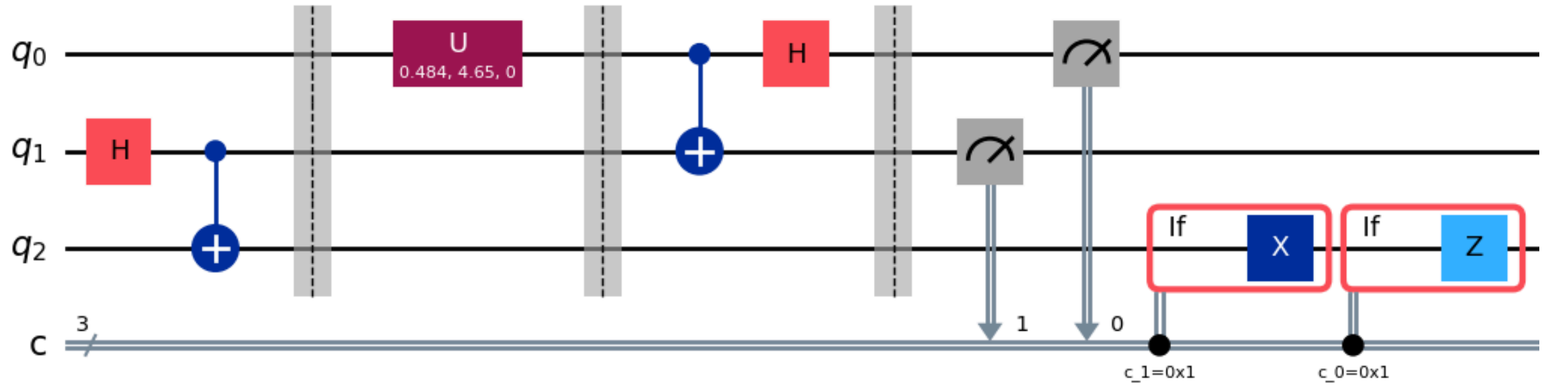
Circuit Diagrams

- Step 3: Entanglement of Alice and Q’s qubit.



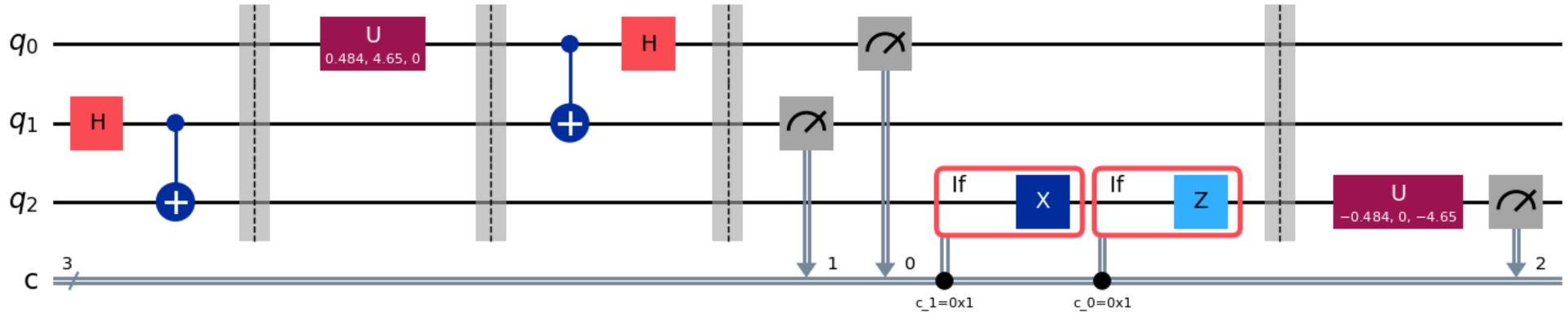
Circuit Diagrams

- Step 4: Measurement by Alice of her and Q’s state.



Circuit Diagrams

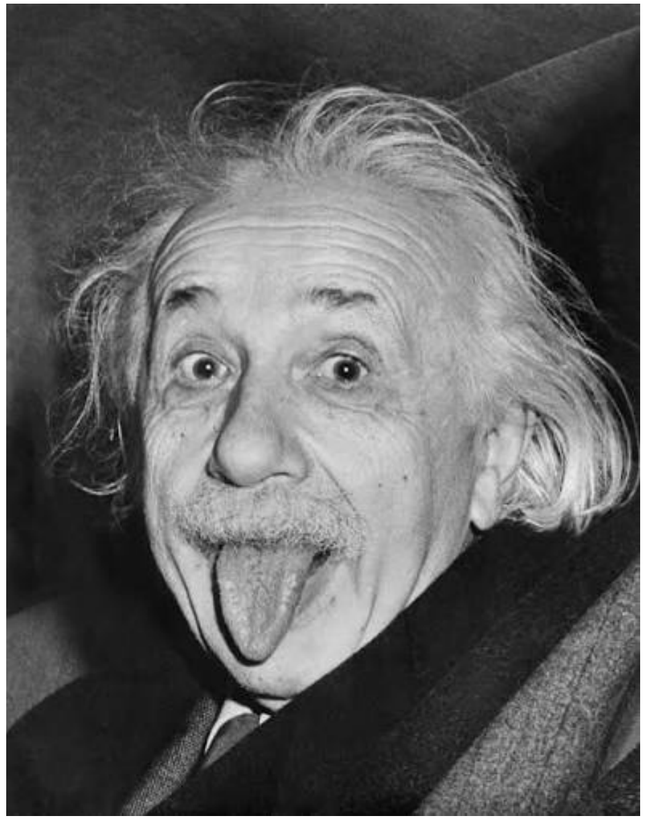
- Step 6: Applying U inverse on Bob’s state to get back the original state of Bob (|0⟩).



But..Wait..Did we break Causality?



But..Wait..Did we break Causality?



NOPE!

Topics to be Covered

1. Classical Information Theory:
   - Building an intuition for certain concepts
   - Compression = Prediction

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2. Quantum Information Theory
   - Quantum analogues of classical information theory

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3. What exactly is ‘Quantum Computing’?
   - 2 common misconceptions about Quantum Computing
   - Building up to a quantum algorithm (Grover’s Algorithm)

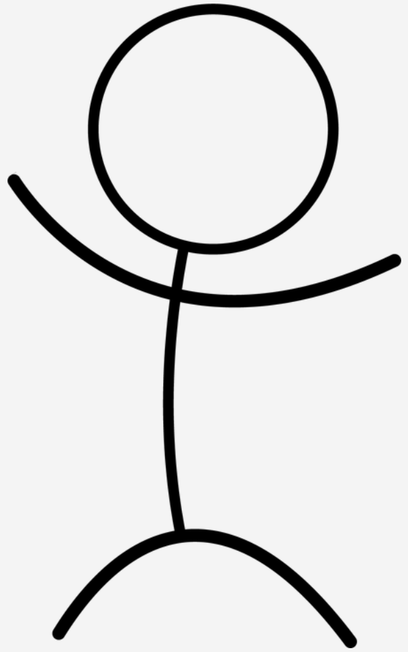
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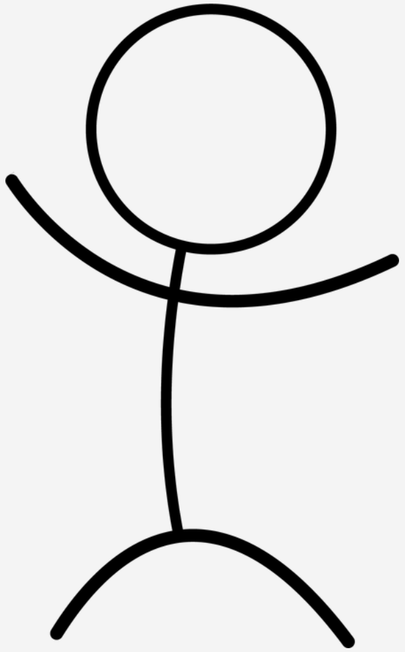
4. Where can we apply Quantum Computing?
   - Quantum Computing is not a be-all end-all
   - Ideal places to apply quantum computing
   - Quantum Computers and Privacy

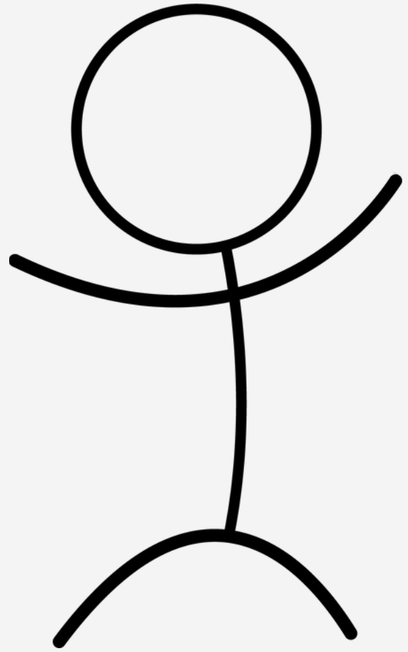
Part I
Classical Information Theory



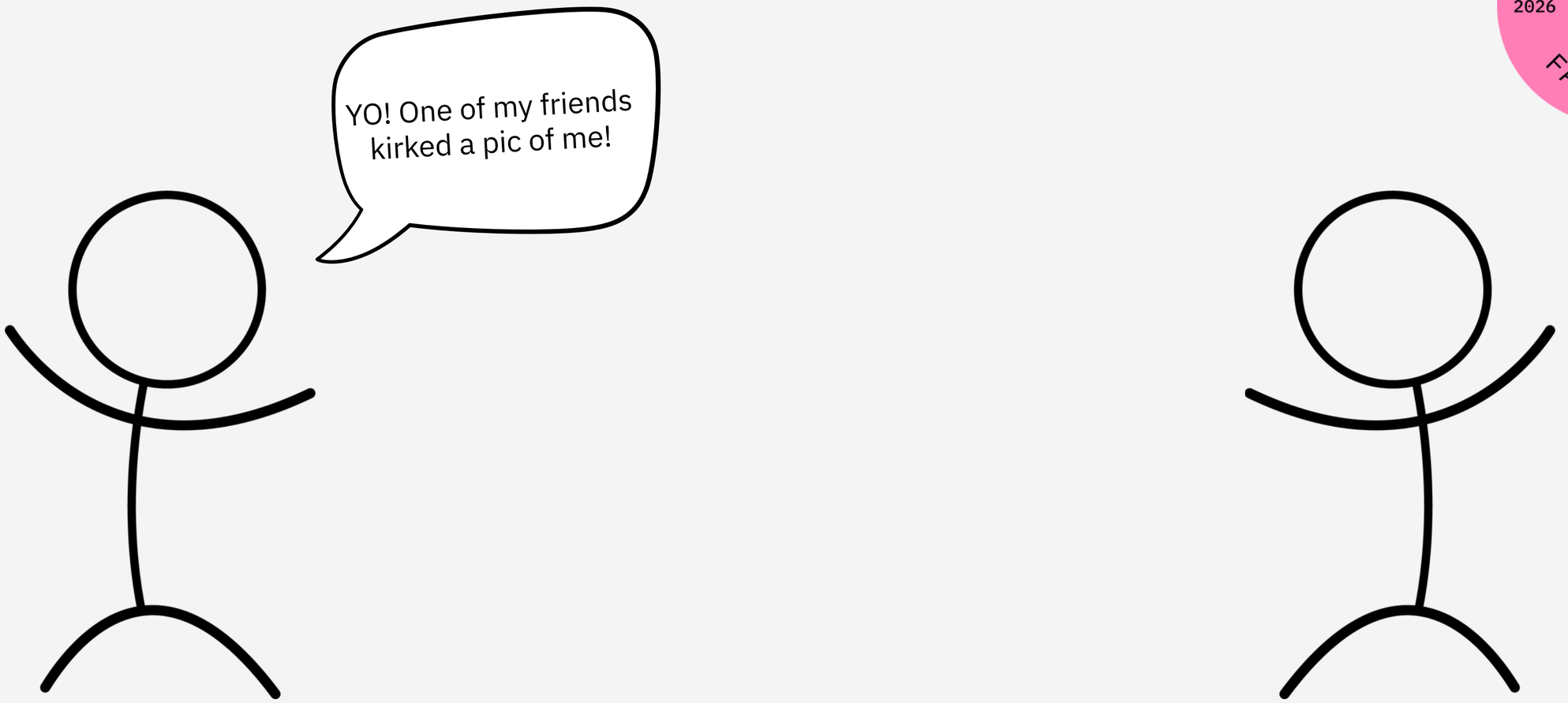

QISKIT
2026
2026
FALL FEST

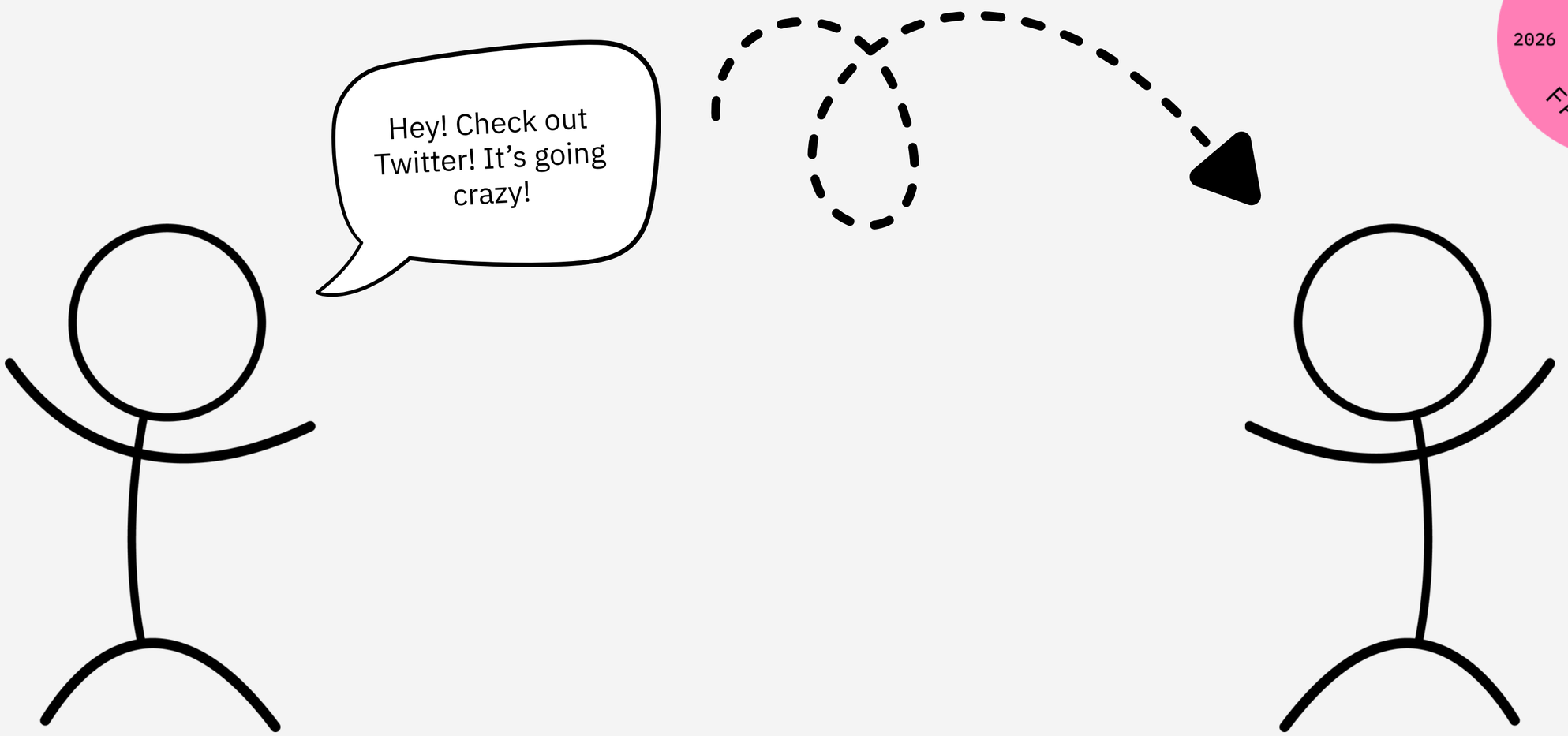






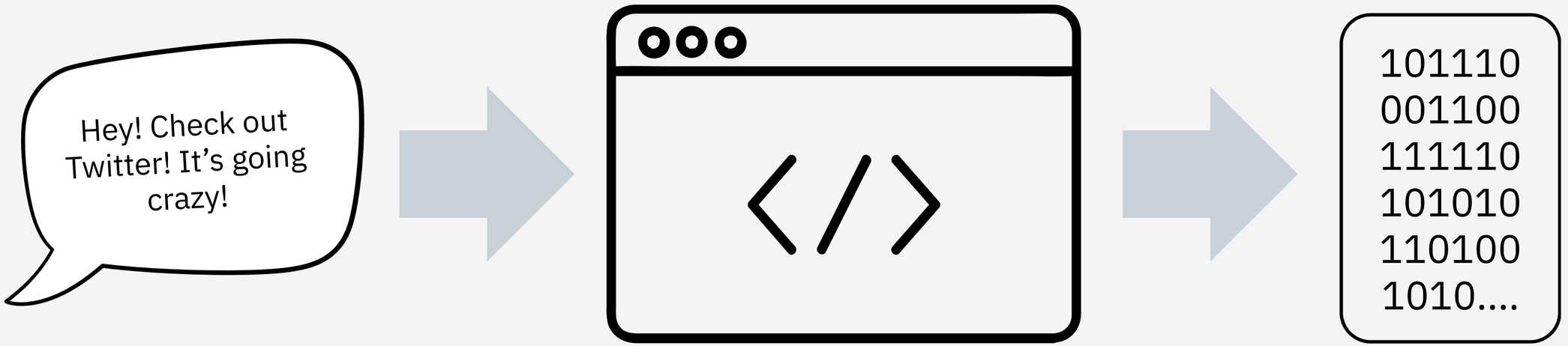



YO! One of my friends kirked a pic of me!


Hey! Check out Twitter! It’s going crazy!


QISKIT
2026
2026
FALL FEST


Hey! Check out Twitter! It’s going crazy!


Hey! Check out Twitter! It’s going crazy!
101110
001100
111110
101010
110100
1010....



101110
001100
111110
101010
110100
1010....

What are some compression algorithms?

ASCII

8 bits/ch

a=01000001
A=01100001
0=00110000

Huffman Coding

~4.2 bits/ch

Neural Methods

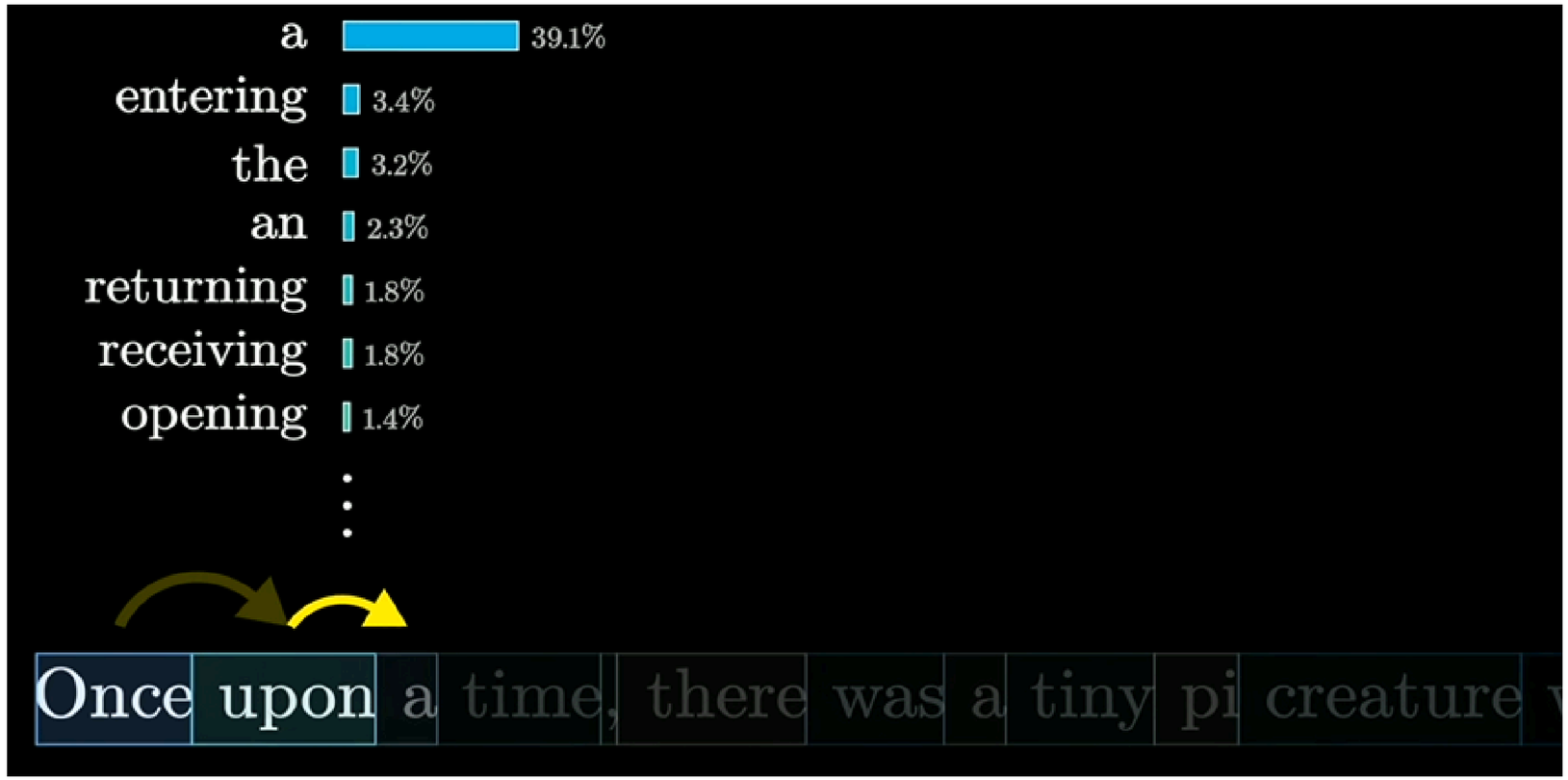
~1.1 bits/ch

Limit?

? bits/ch



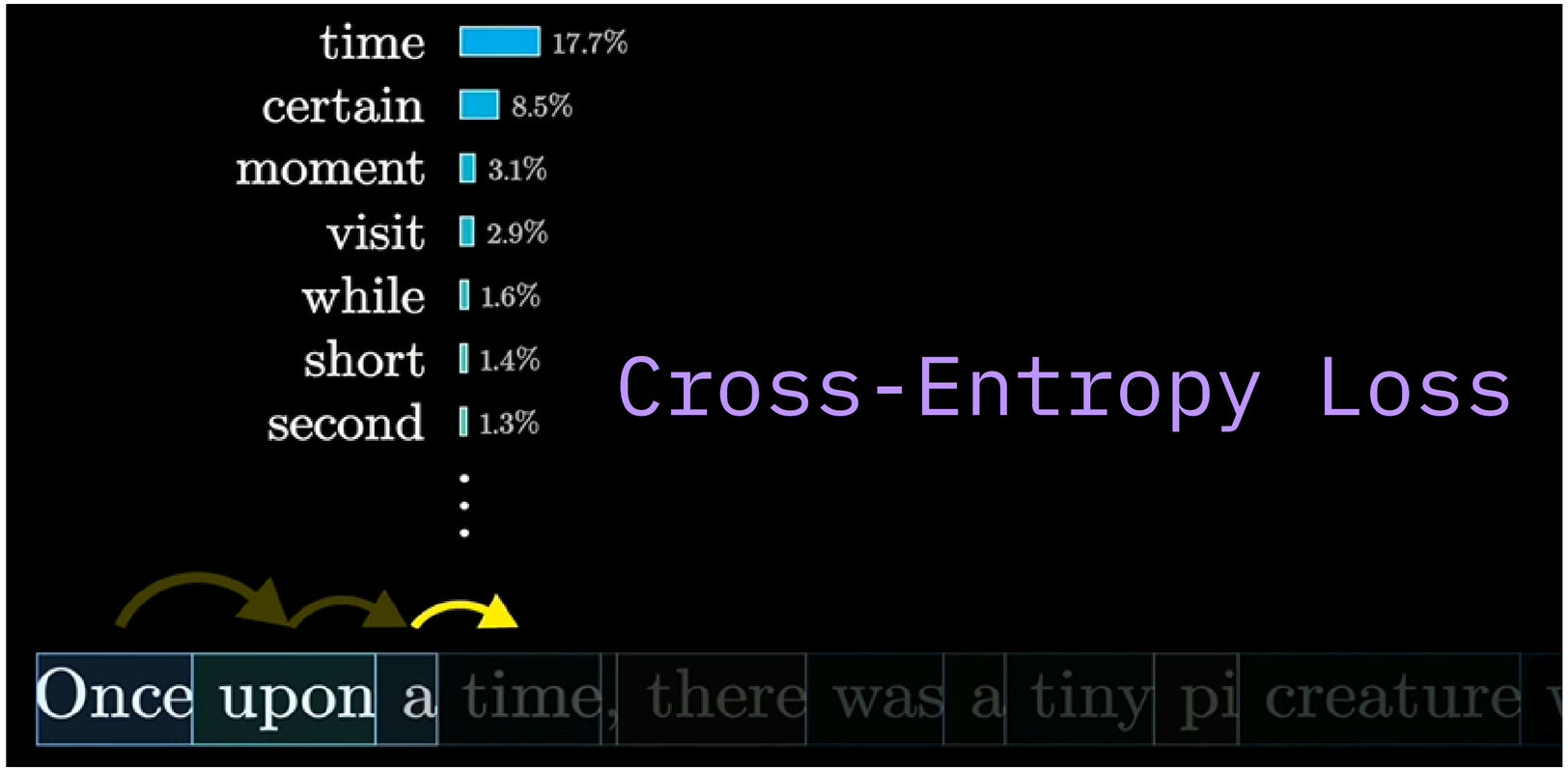
Slight Detour: What do LLMs do?



Slight Detour: What do LLMs do?



Slight Detour: What do LLMs do?

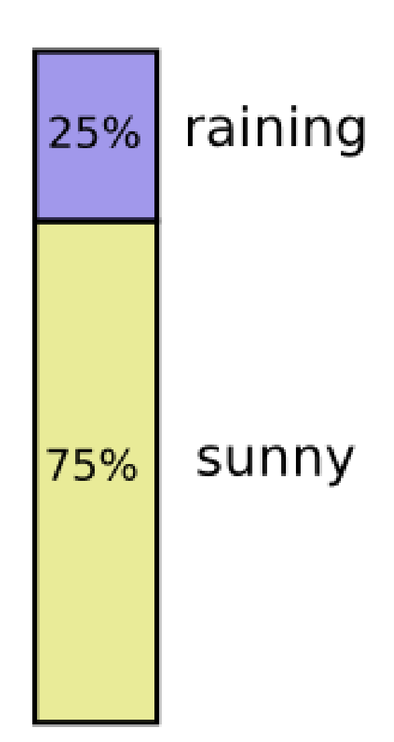


Compression is equivalent to Prediction*

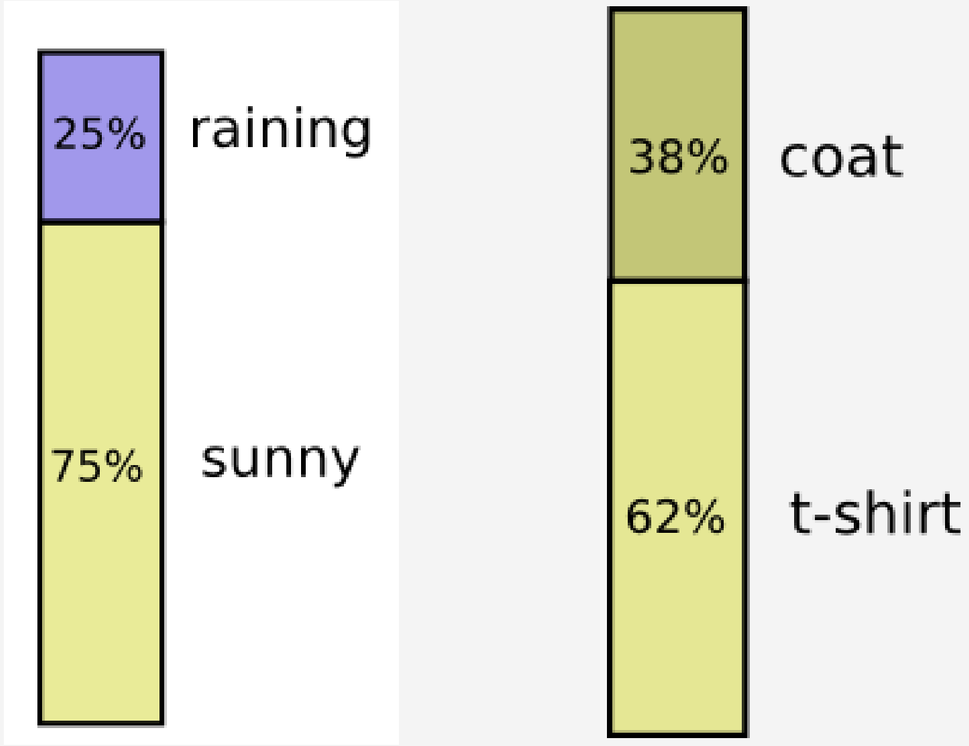


*Terms and Conditions apply

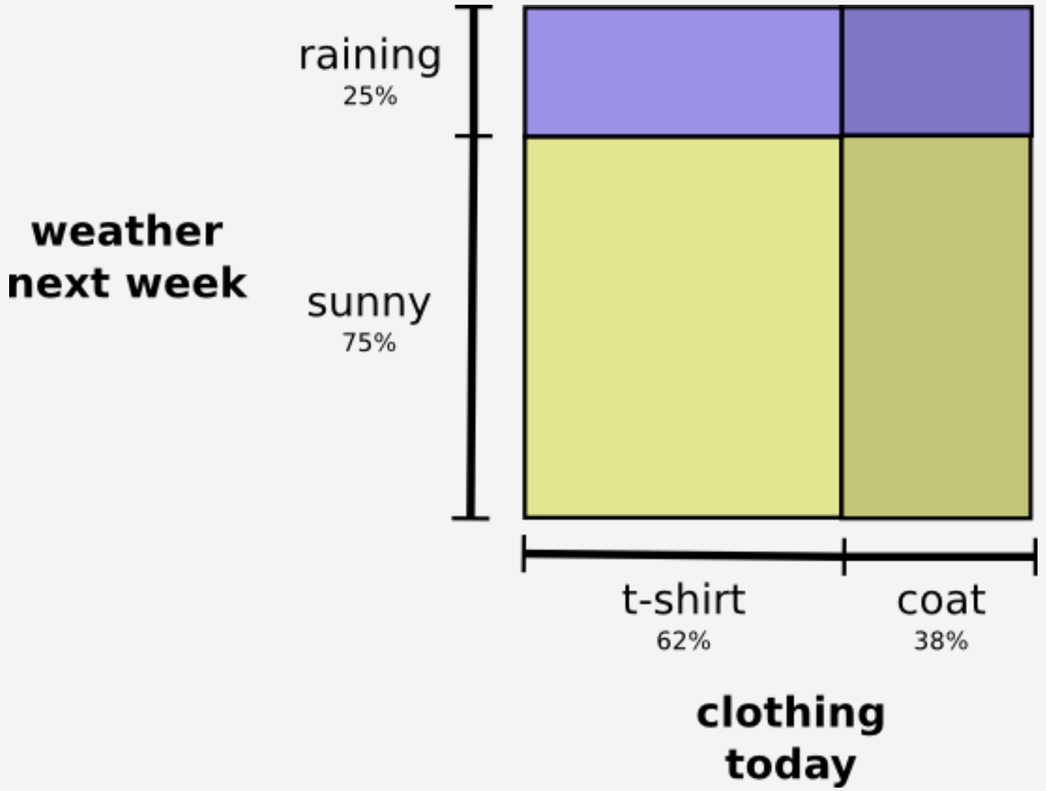
Visualising Probability Distributions



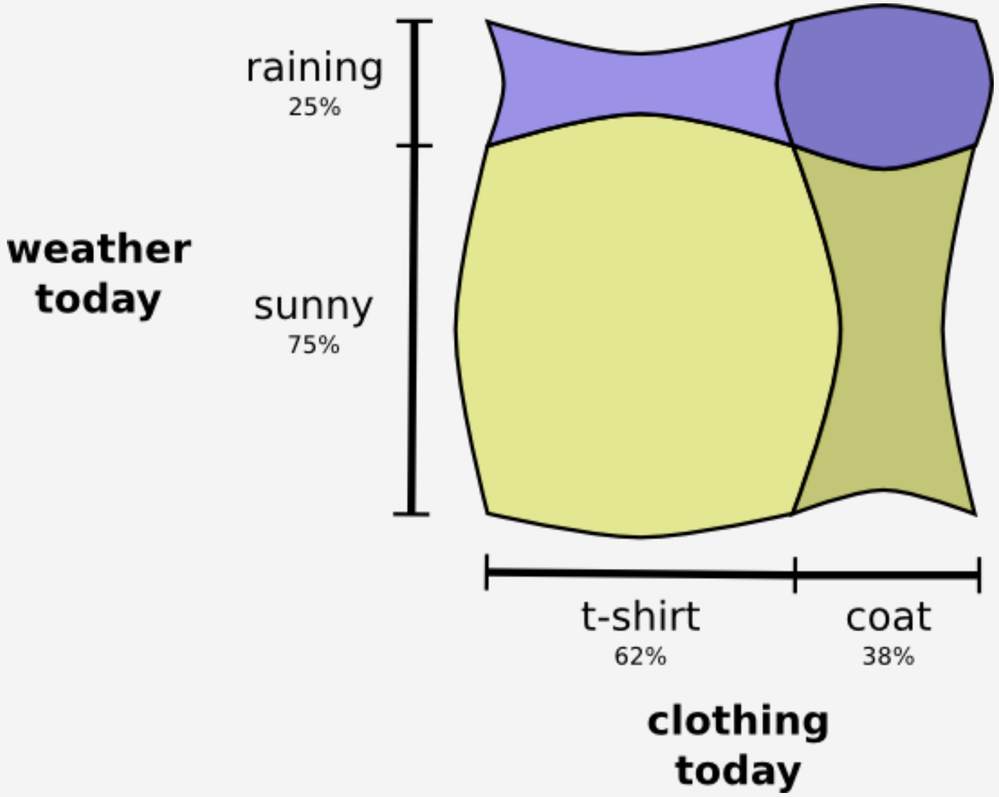
Visualising Probability Distributions



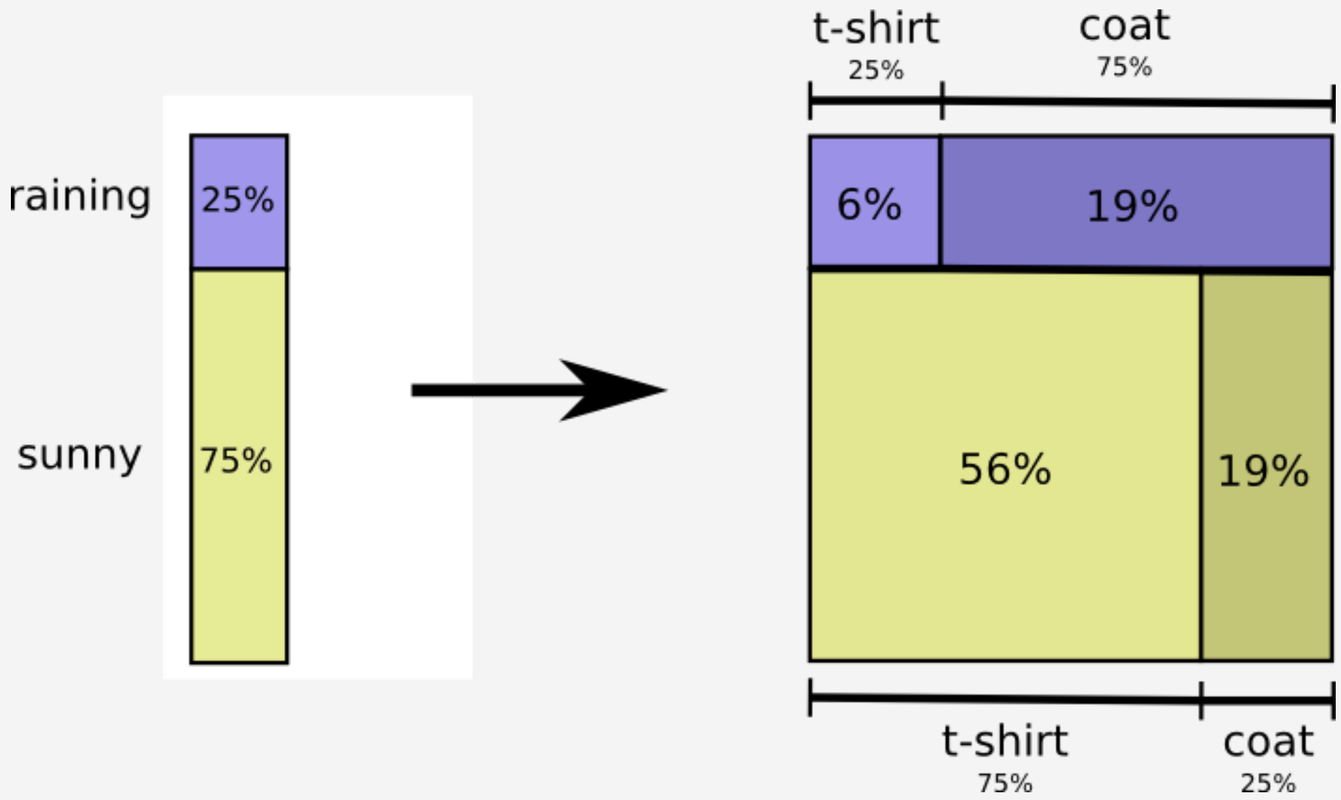
Visualising Probability Distributions



Visualising Probability Distributions

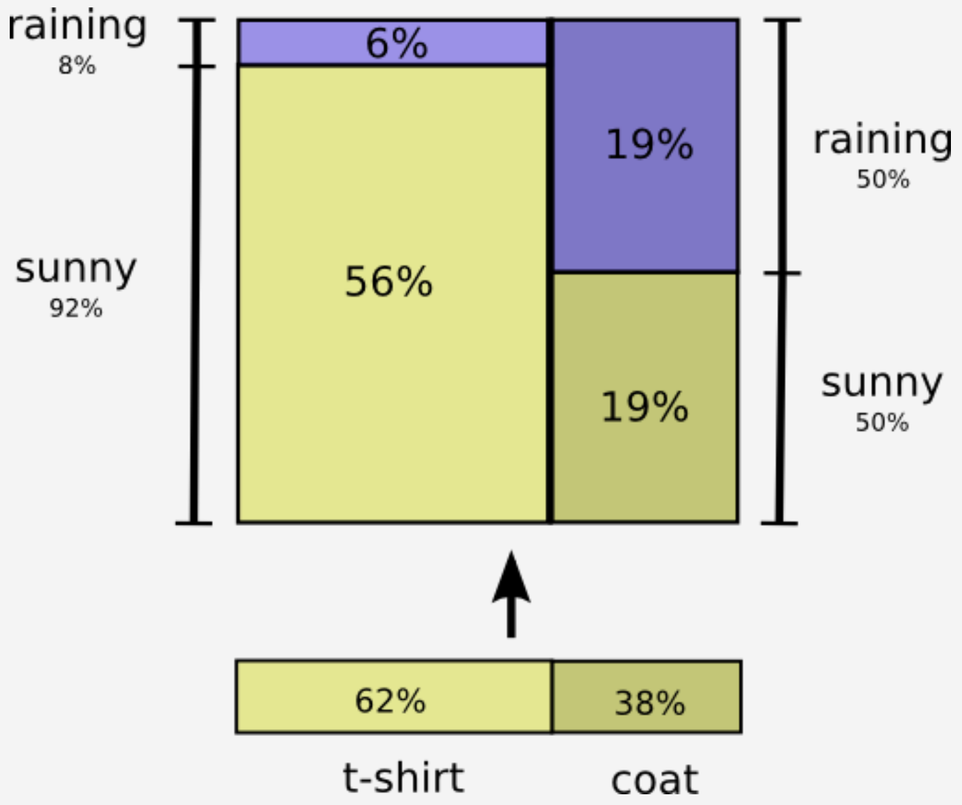


Visualising Probability Distributions

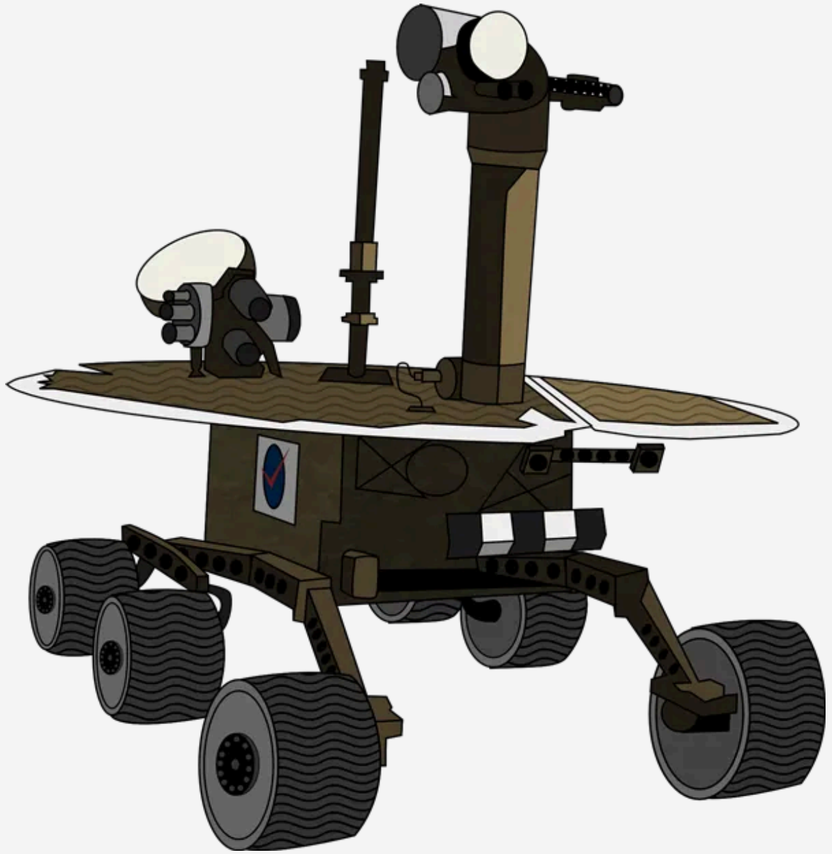


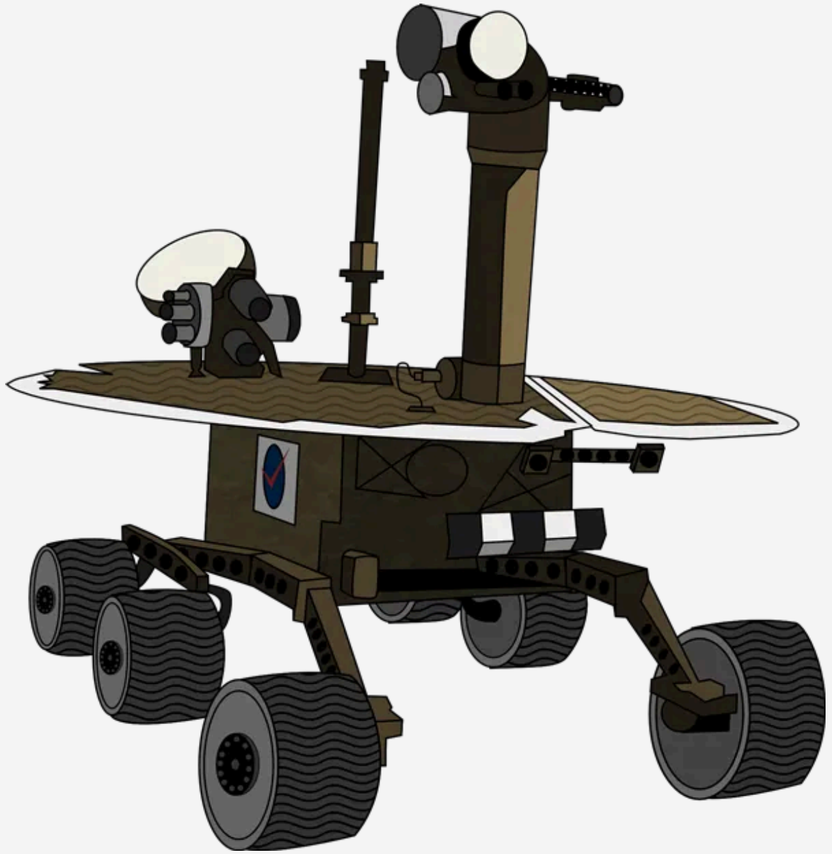
$$p(\text{rain},\text{coat}) = p(\text{rain}) \cdot p(\text{coat} \mid \text{rain})$$

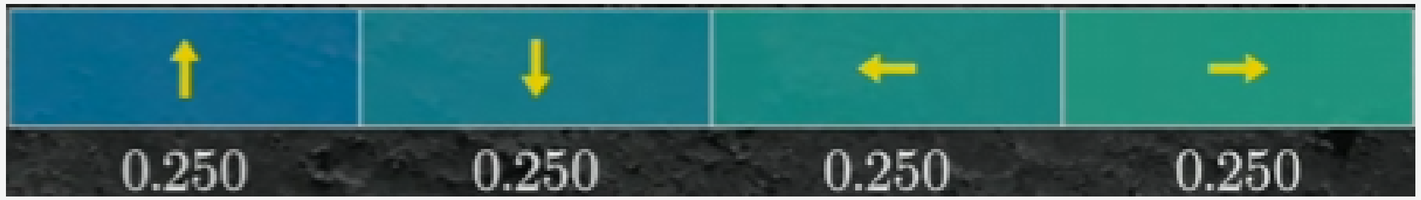
Visualising Probability Distributions

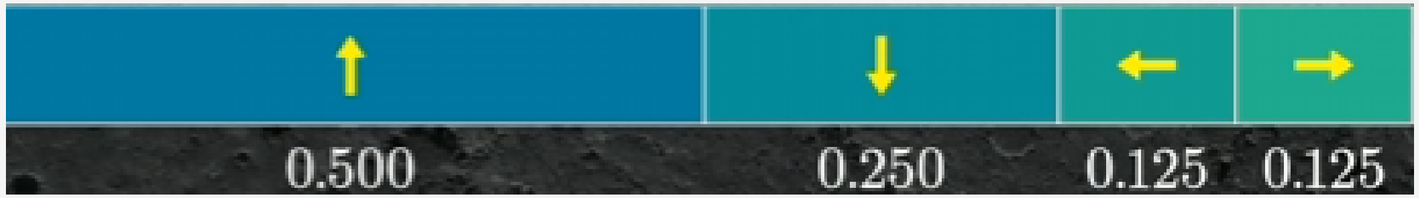


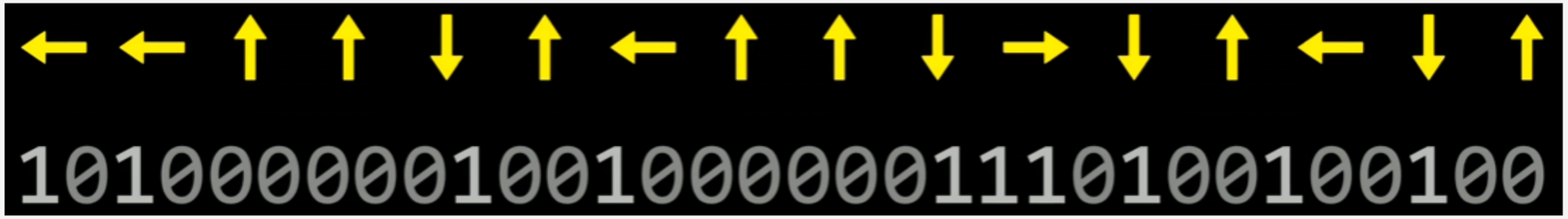
$$p(\text{rain},\text{coat}) = p(\text{coat}) \cdot p(\text{rain} \mid \text{coat})$$





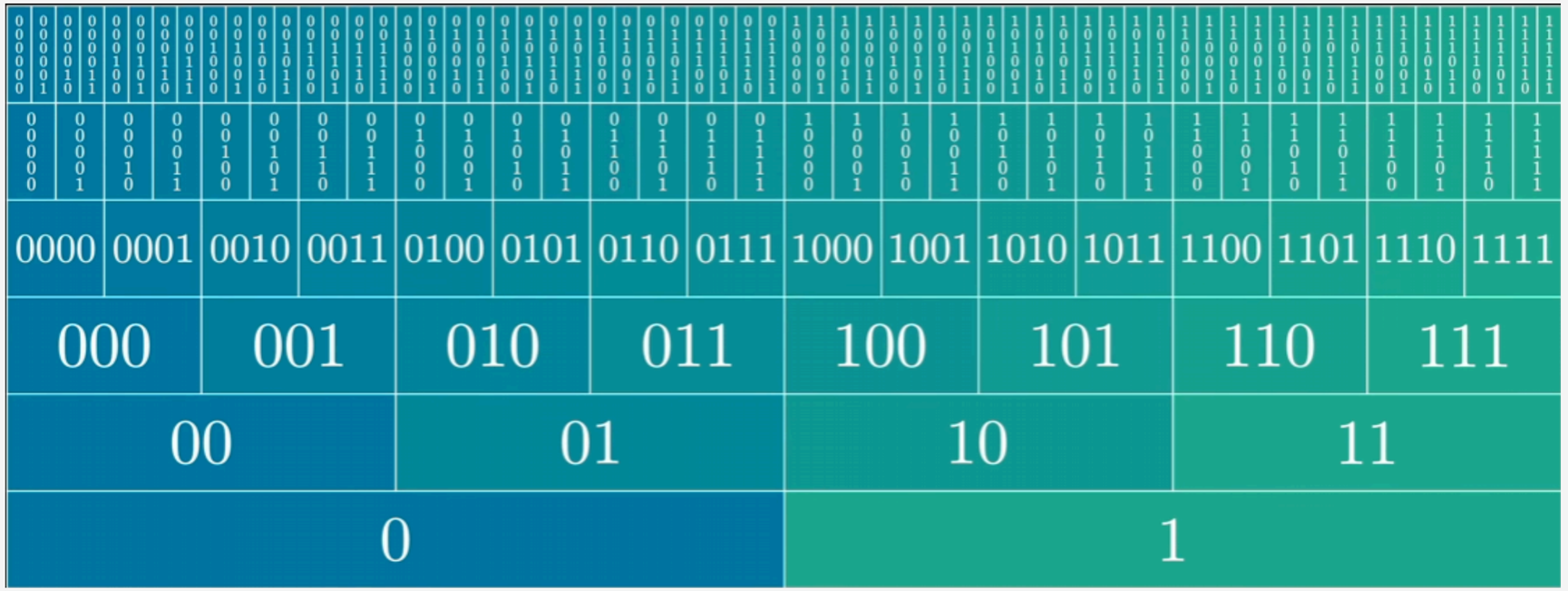

0.250
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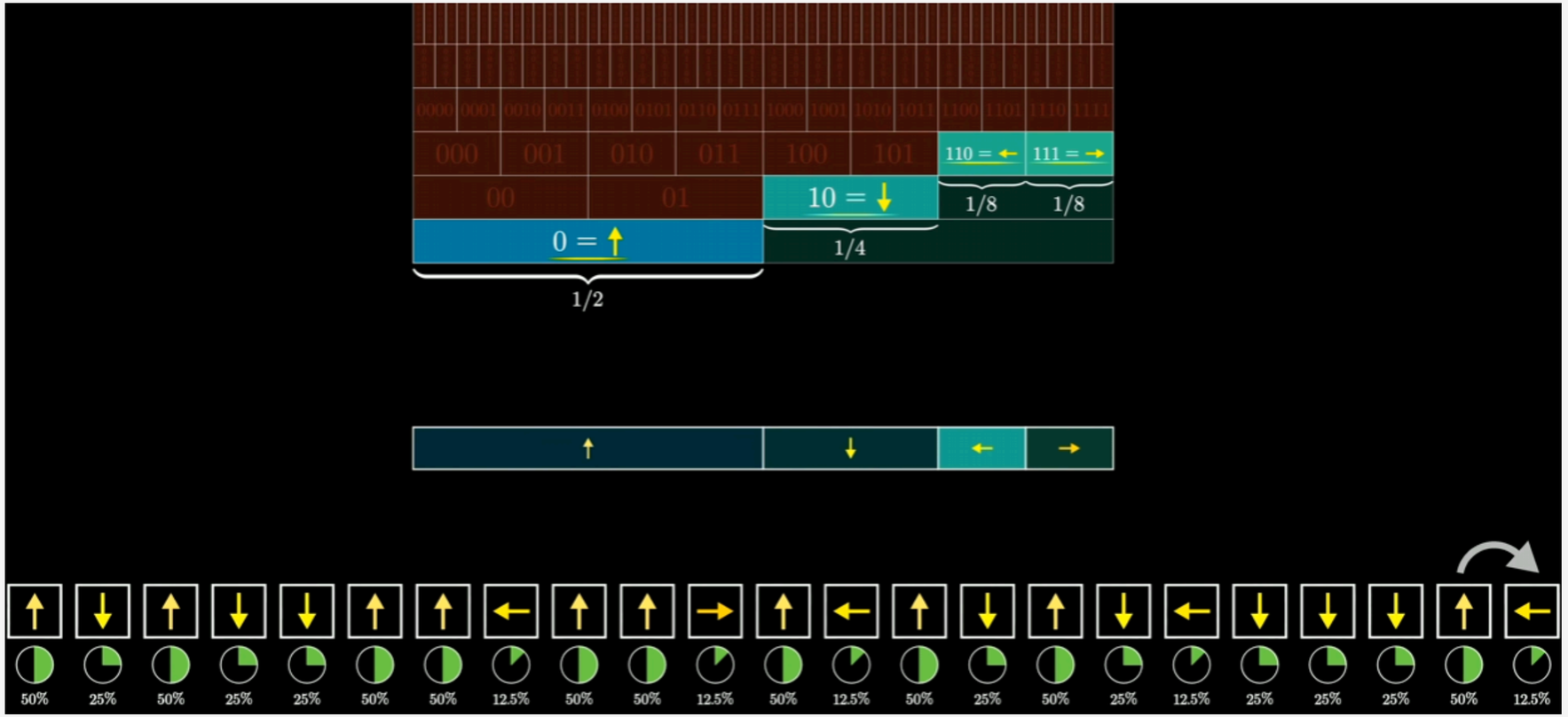

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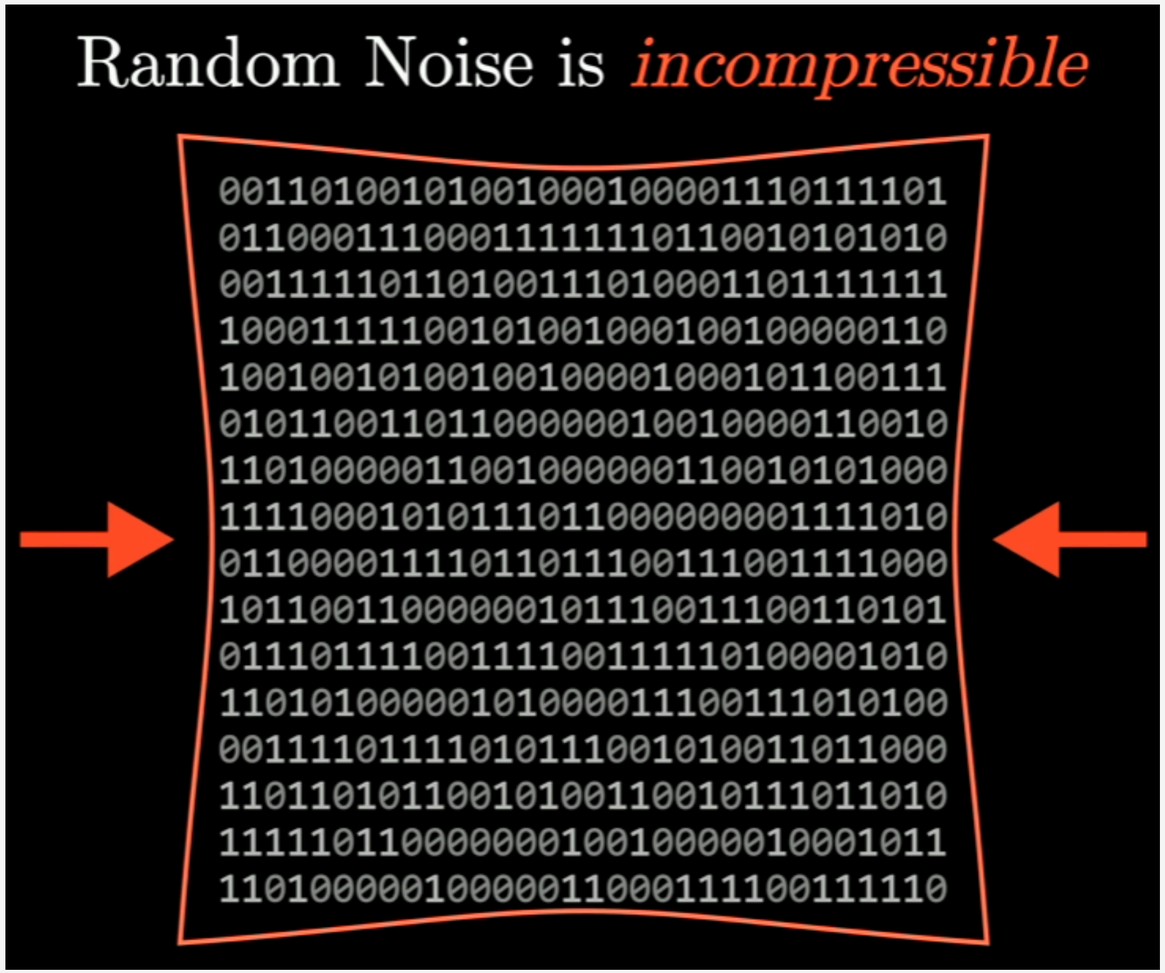

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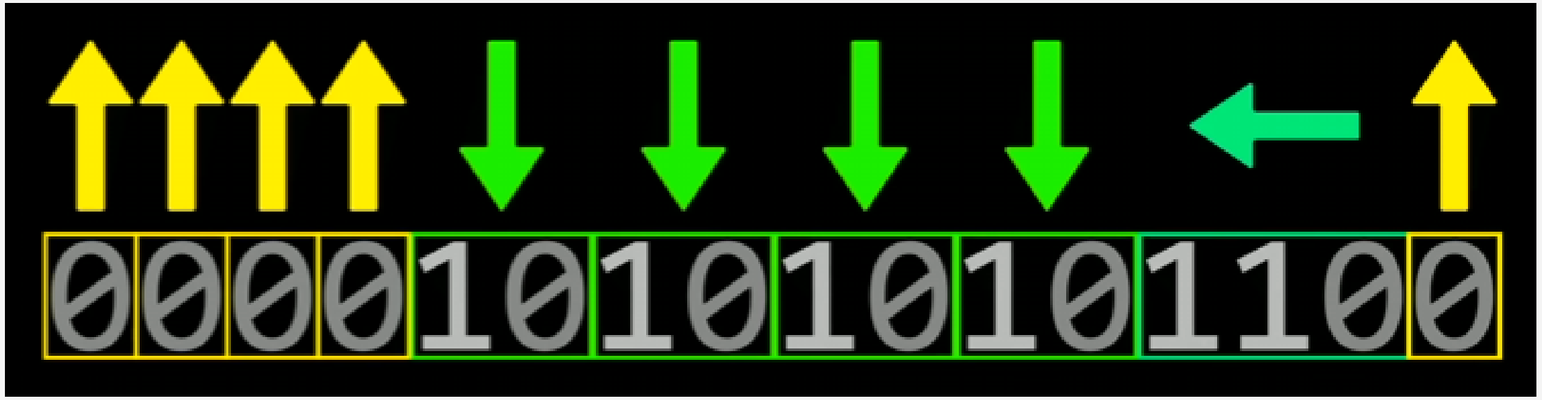


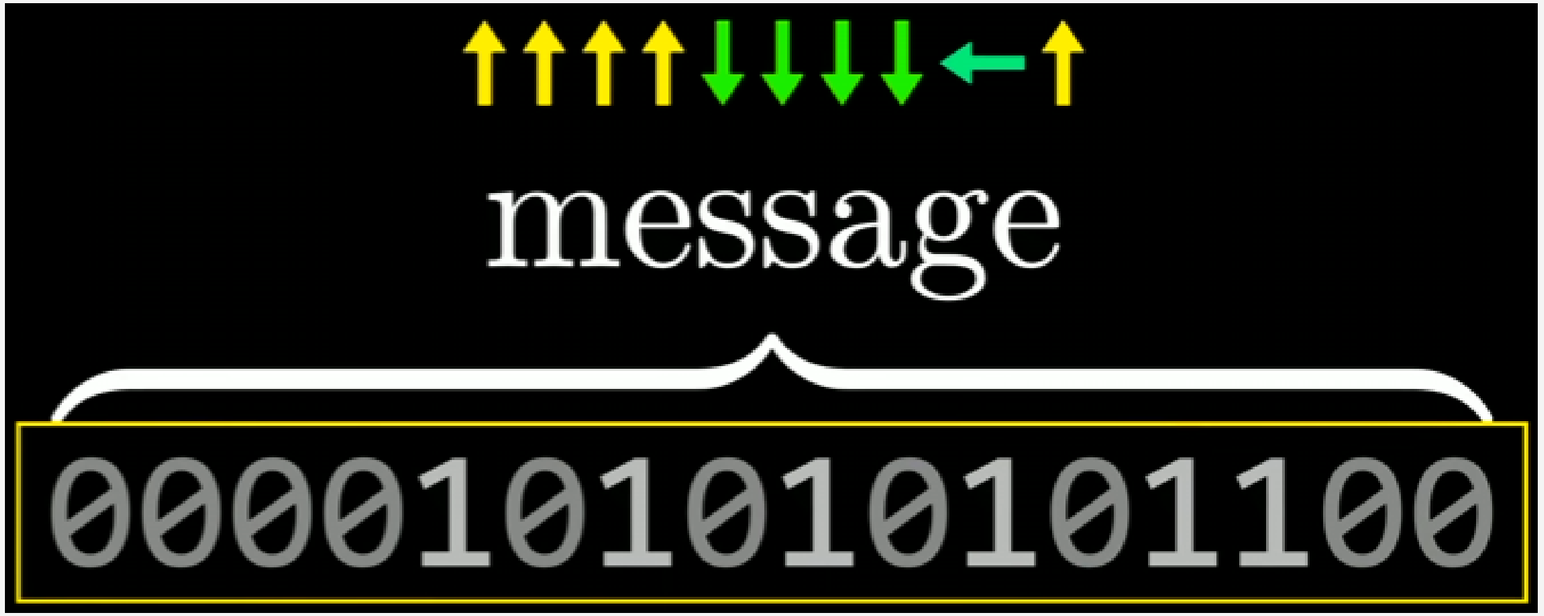
BOARD TIME!!!


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000 001 010 011 100 101 110 111
00 01 10 11
0
1

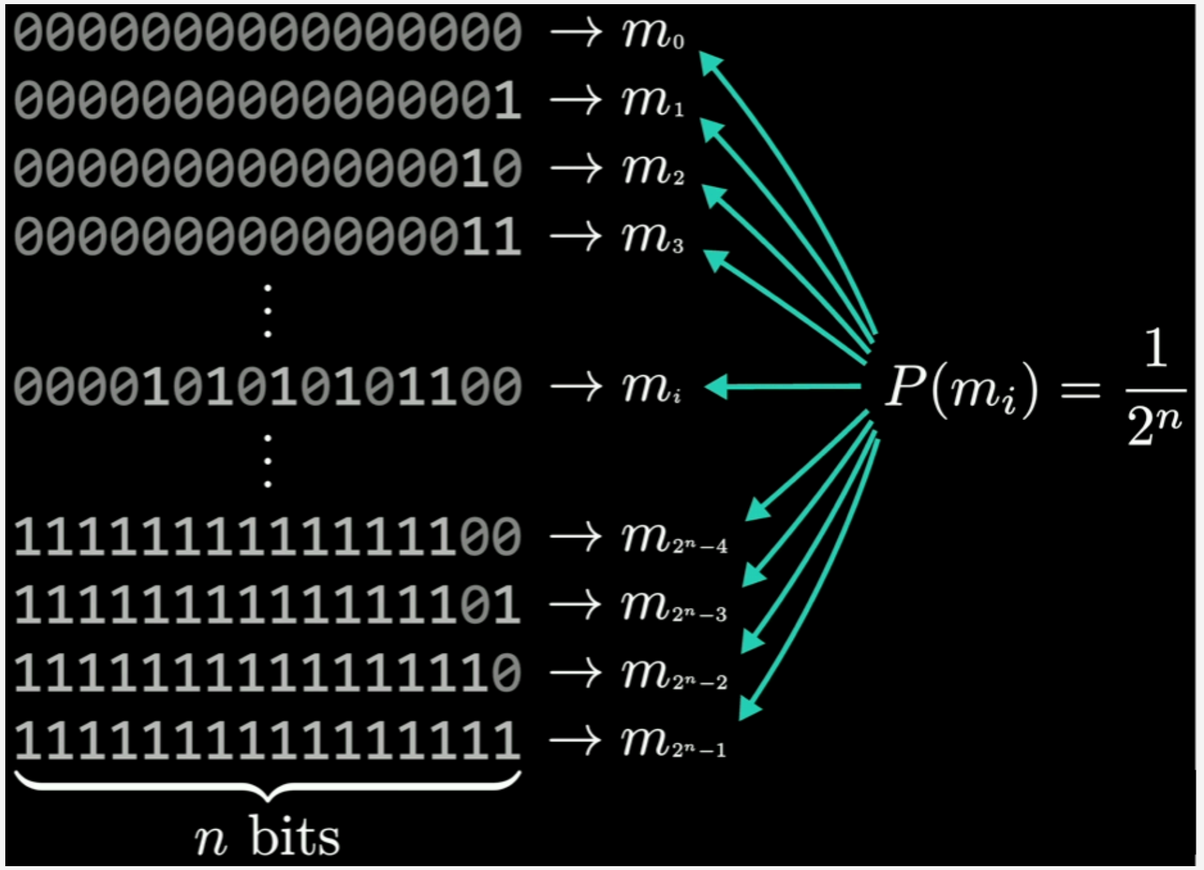

0000 0001 0010 0011 0100 0101 0110 0111 1000 1001 1010 1011 1100 1101 1110 1111
000 001 010 011 100 101 110 = ← 111 = →
00 01 10 = ↓ 1/8 1/8
0 = ↑ 1/4
1/2
50% 25% 50% 25% 25% 50% 50% 12.5% 50% 50% 12.5% 50% 12.5% 50% 25% 50% 25% 12.5% 25% 25% 25% 50% 12.5%


Random Noise is *incompressible*
00110100101001000100001110111101
01100011100011111110110010101010
00111110110100111010001101111111
10001111100101001000100100000110
10010010100100100001000101100111
01011001101100000010010000110010
11010000011001000000110010101000
11110001010111011000000001111010
01100001111011011100111001111000
10110011000000101110011100110101
01110111100111100111110100001010
11010100000101000011100111010100
00111101111010111001010011011000
11011010110010100110010111011010
11111011000000010010000010001011
11010000010000011000111100111110


0000101010101100

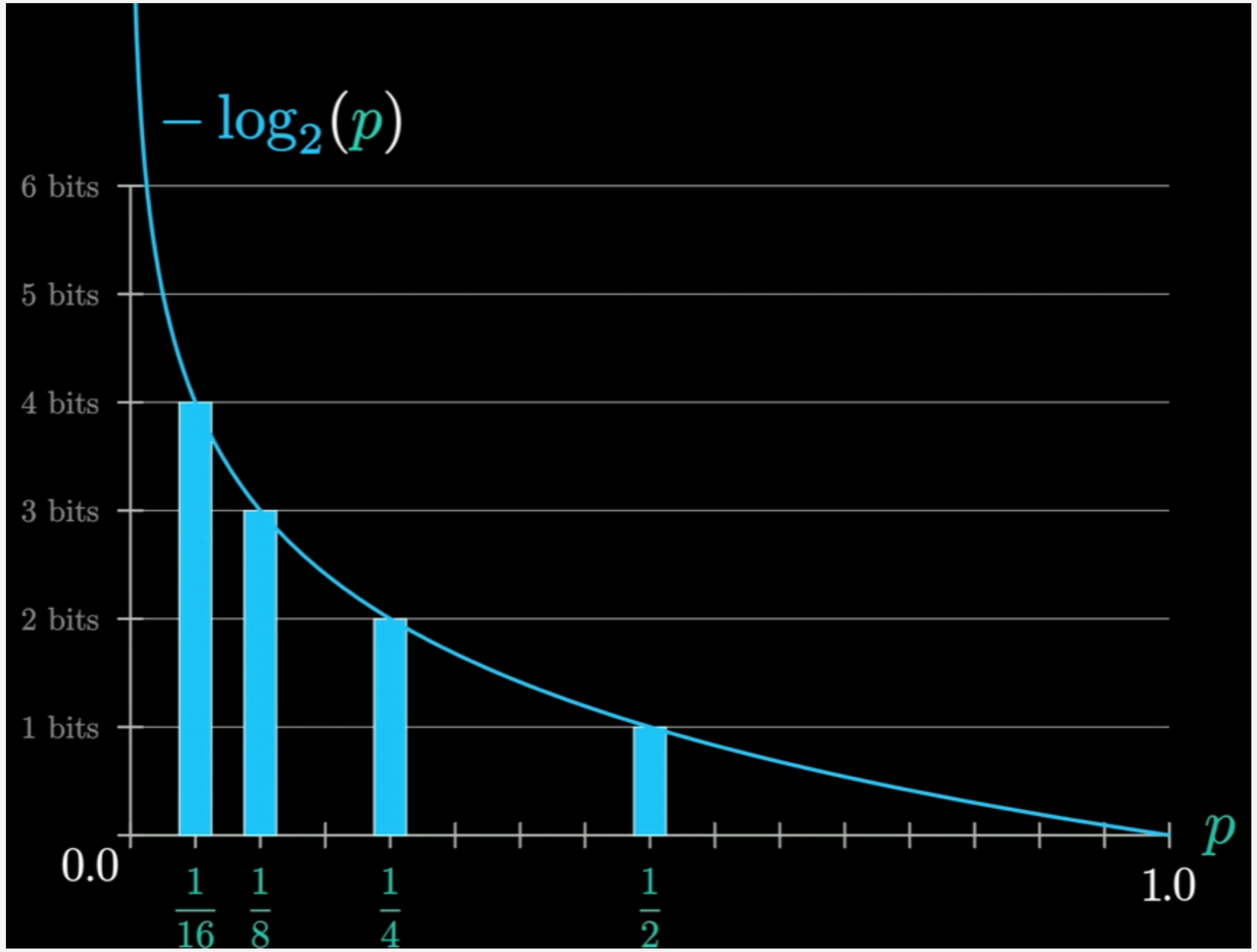

message
0000101010101100

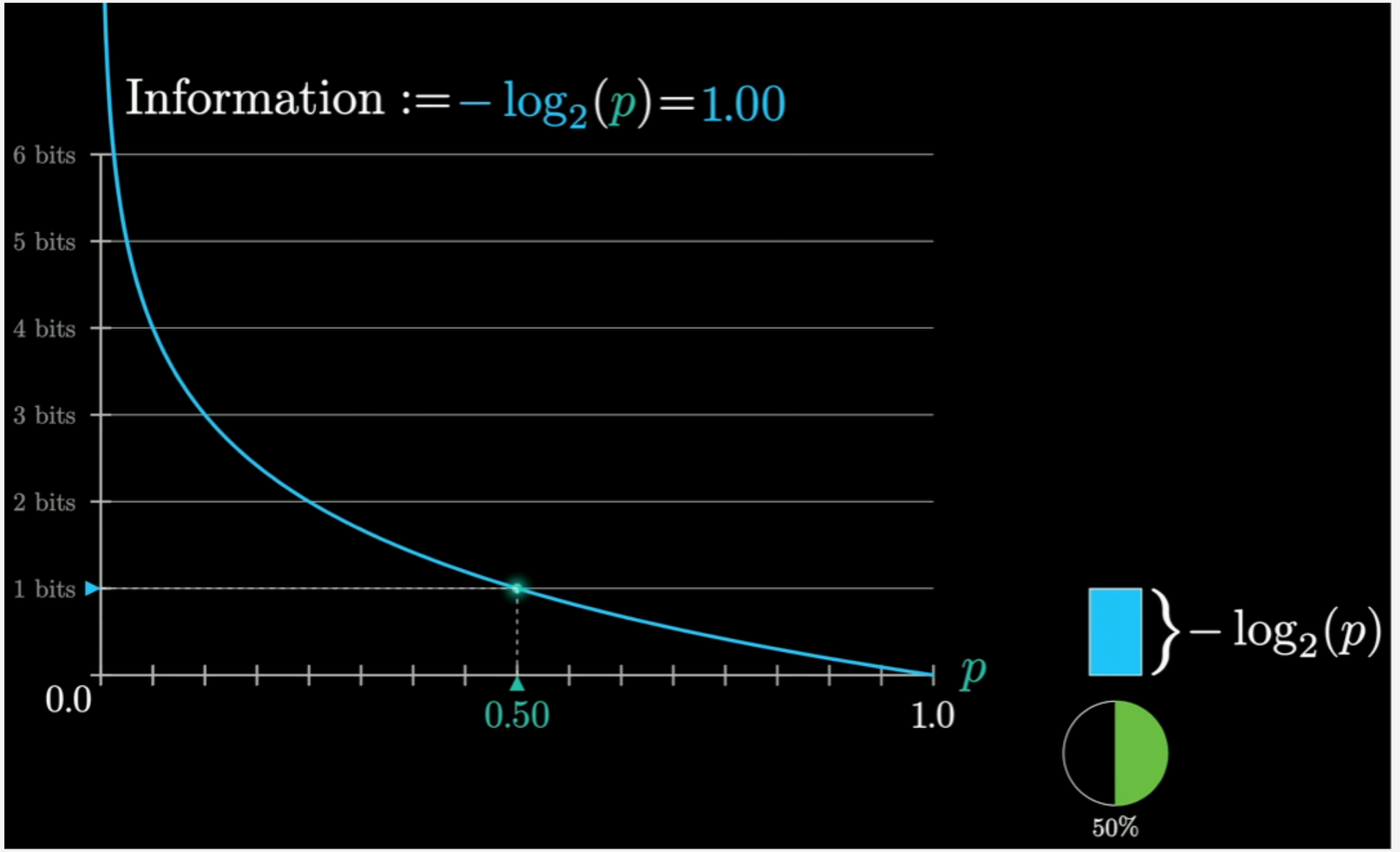

0000000000000000 → m_0
0000000000000001 → m_1
0000000000000010 → m_2
0000000000000011 → m_3
0000101010101100 → m_i
1111111111111100 → m_{2^n-4}
1111111111111101 → m_{2^n-3}
1111111111111110 → m_{2^n-2}
1111111111111111 → m_{2^n-1}
2^n
n bits

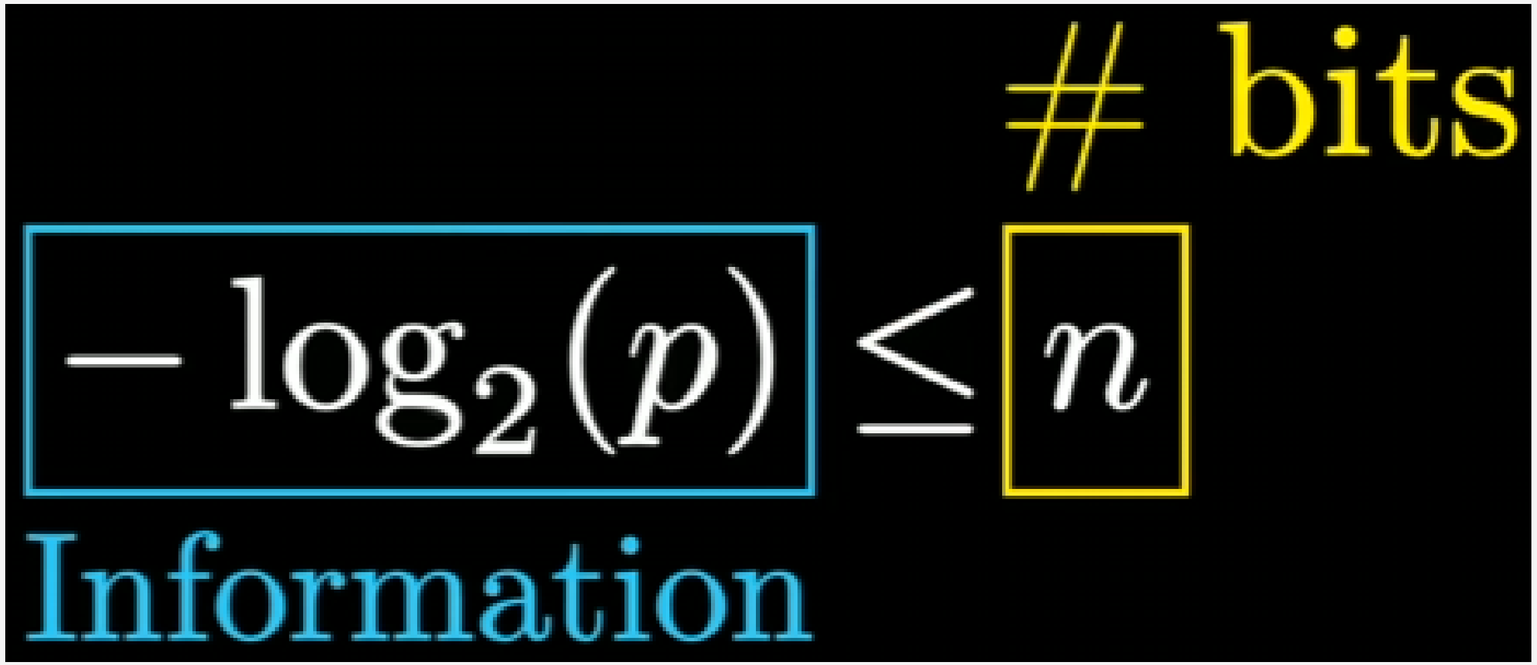

0000000000000000 → m_0
0000000000000001 → m_1
0000000000000010 → m_2
0000000000000011 → m_3
0000101010101100 → m_i
1111111111111100 → m_{2^n-4}
1111111111111101 → m_{2^n-3}
1111111111111110 → m_{2^n-2}
1111111111111111 → m_{2^n-1}
n bits
$P(m_i) = \frac{1}{2^n}$



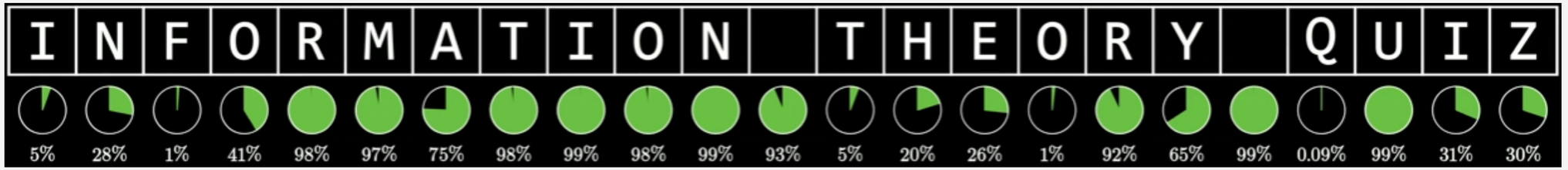
BACK TO THE BOARD

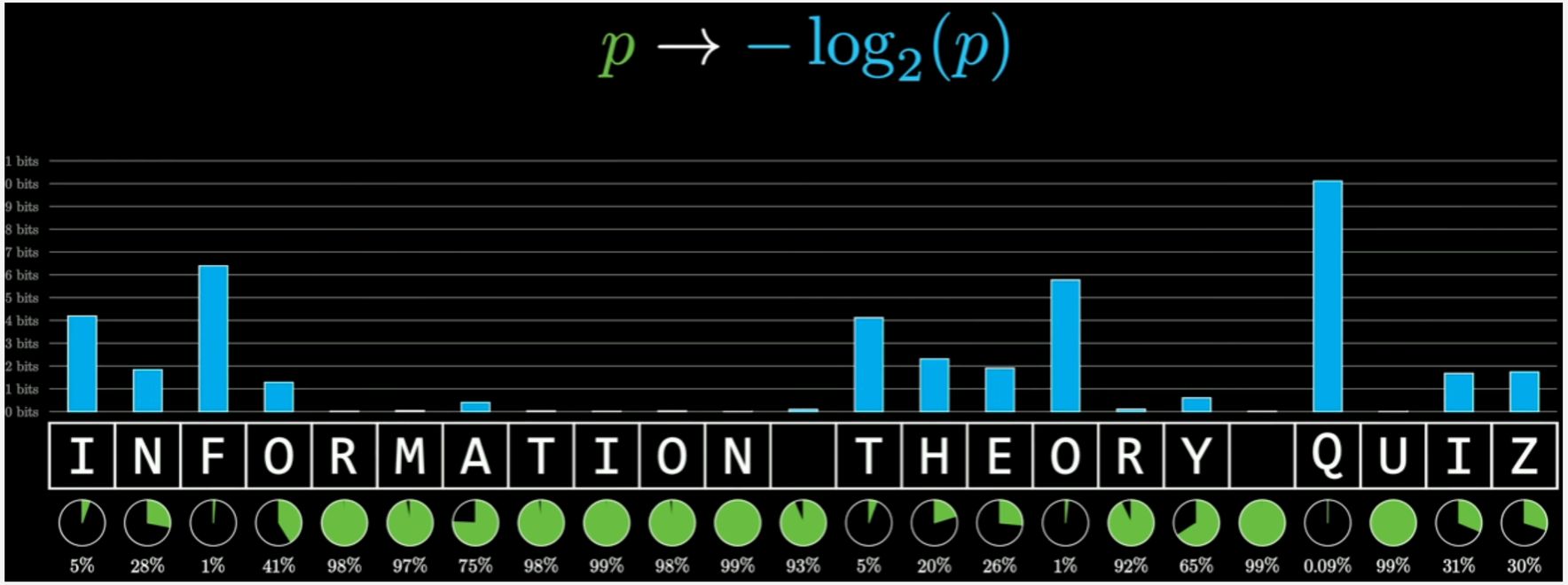

$-\log_2(p)$
6 bits
5 bits
4 bits
3 bits
2 bits
1 bits
0.0
$\frac{1}{16}$
$\frac{1}{8}$
$\frac{1}{4}$
$\frac{1}{2}$
1.0
p

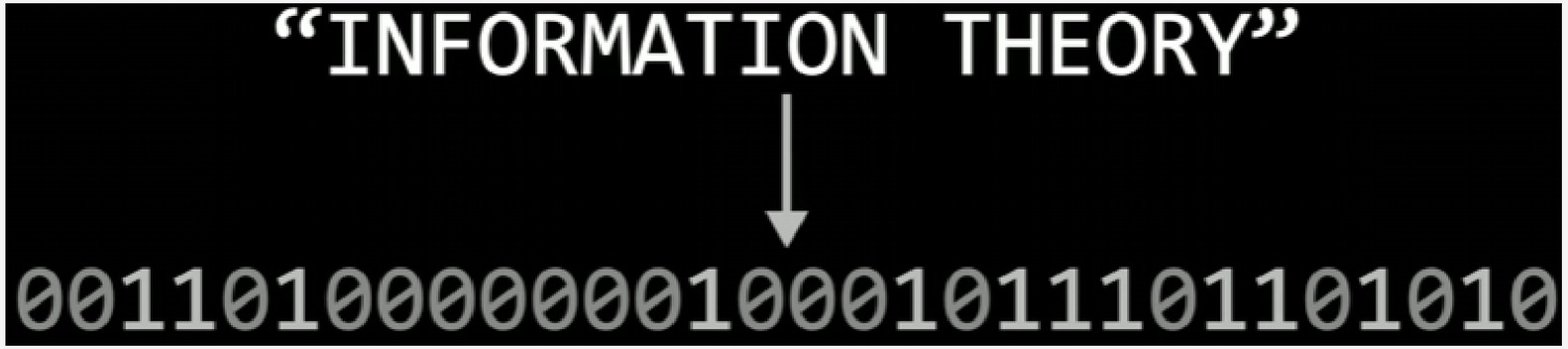

Information := $-\log_2(p) = 1.00$
6 bits
5 bits
4 bits
3 bits
2 bits
1 bits
0.0
0.50
1.0
p
$-\log_2(p)$
50%



$$\overline{-\log_2(p)} \leq \overline{\#\ \text{bits}}$$


INFORMATION THEORY QUIZ
5%
28%
1%
41%
98%
97%
75%
98%
99%
98%
99%
93%
5%
20%
26%
1%
92%
65%
99%
0.09%
99%
31%
30%


$p \to -\log_2(p)$
1 bits
0 bits
9 bits
8 bits
7 bits
6 bits
5 bits
4 bits
3 bits
2 bits
1 bits
0 bits
I N F O R M A T I O N T H E O R Y Q U I Z
5% 28% 1% 41% 98% 97% 75% 98% 99% 98% 99% 93% 5% 20% 26% 1% 92% 65% 99% 0.09% 99% 31% 30%


“INFORMATION THEORY”
00110100000001000101110110101010

$$P(\text{“INFORMATION THEORY”})$$

$$\Vert$$

$$p_0 \cdot p_1 \cdot p_2 \cdot p_3 \cdots p_n$$

$$-\log_2\left(P(\text{“INFORMATION THEORY”})\right)$$

$$\Vert$$

$$-\log_2\left(p_0 \cdot p_1 \cdot p_2 \cdot p_3 \cdots p_n\right)$$

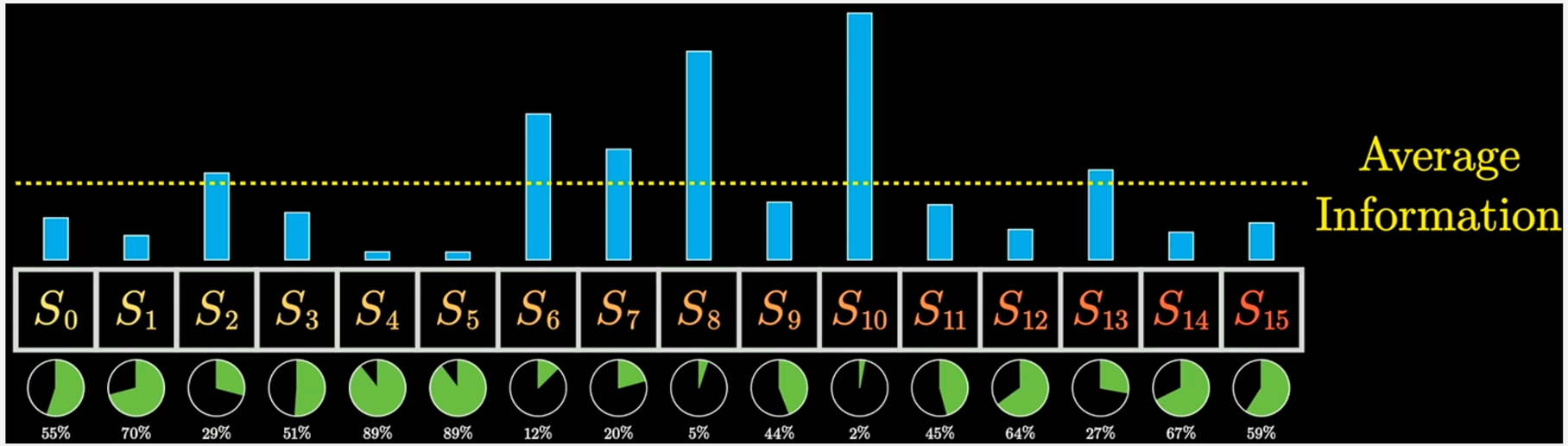
Optimal Encoding length

$\approx$

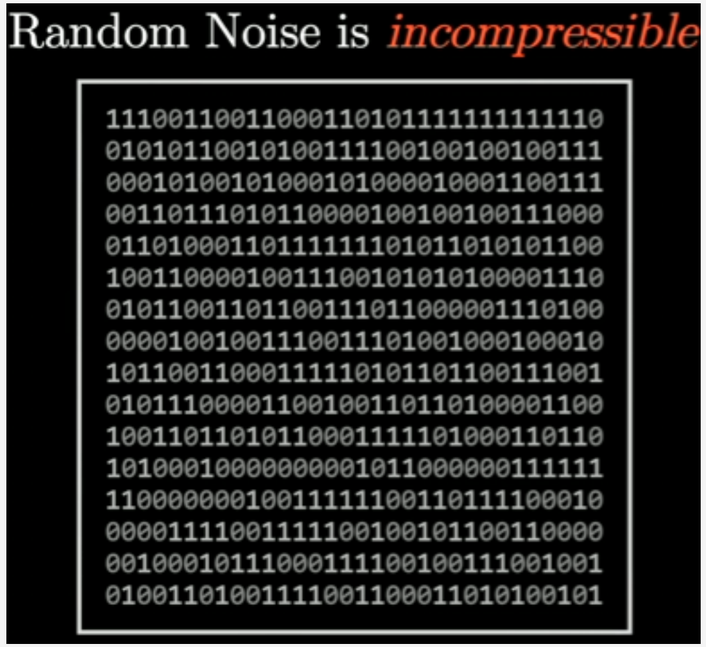
Total information

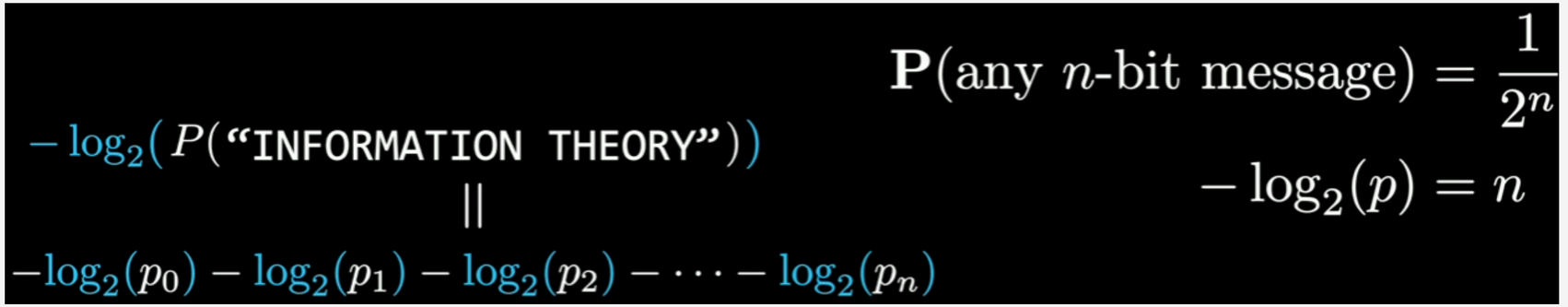
$$-\log_2(p_0) - \log_2(p_1) - \log_2(p_2) - \cdots - \log_2(p_n)$$

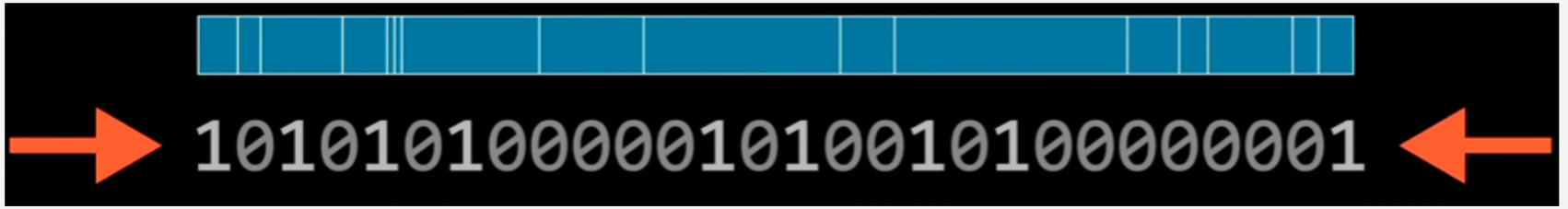


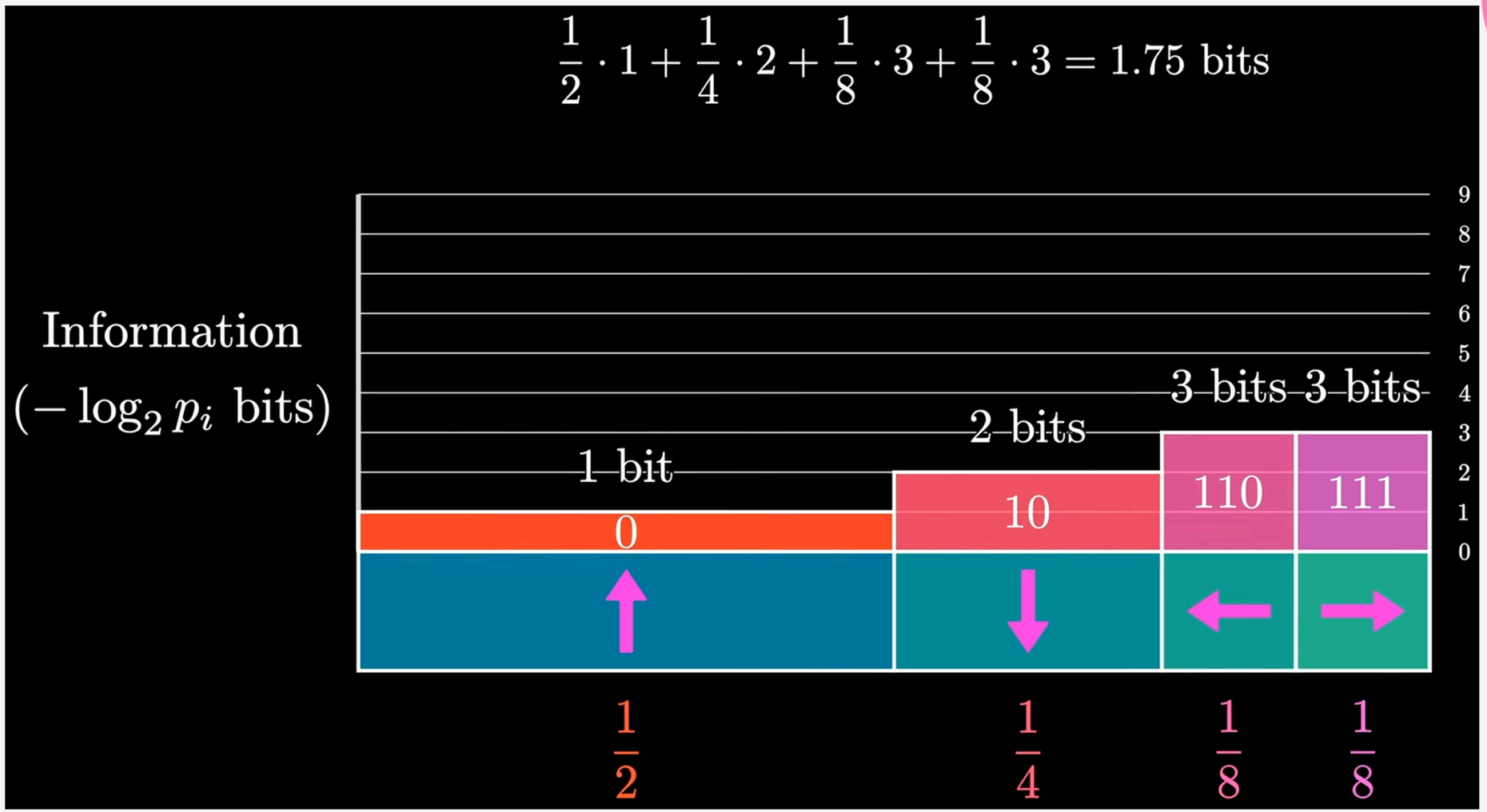

Average Information
S_0 S_1 S_2 S_3 S_4 S_5 S_6 S_7 S_8 S_9 S_{10} S_{11} S_{12} S_{13} S_{14} S_{15}
55% 70% 29% 51% 89% 89% 12% 20% 5% 44% 2% 45% 64% 27% 67% 59%

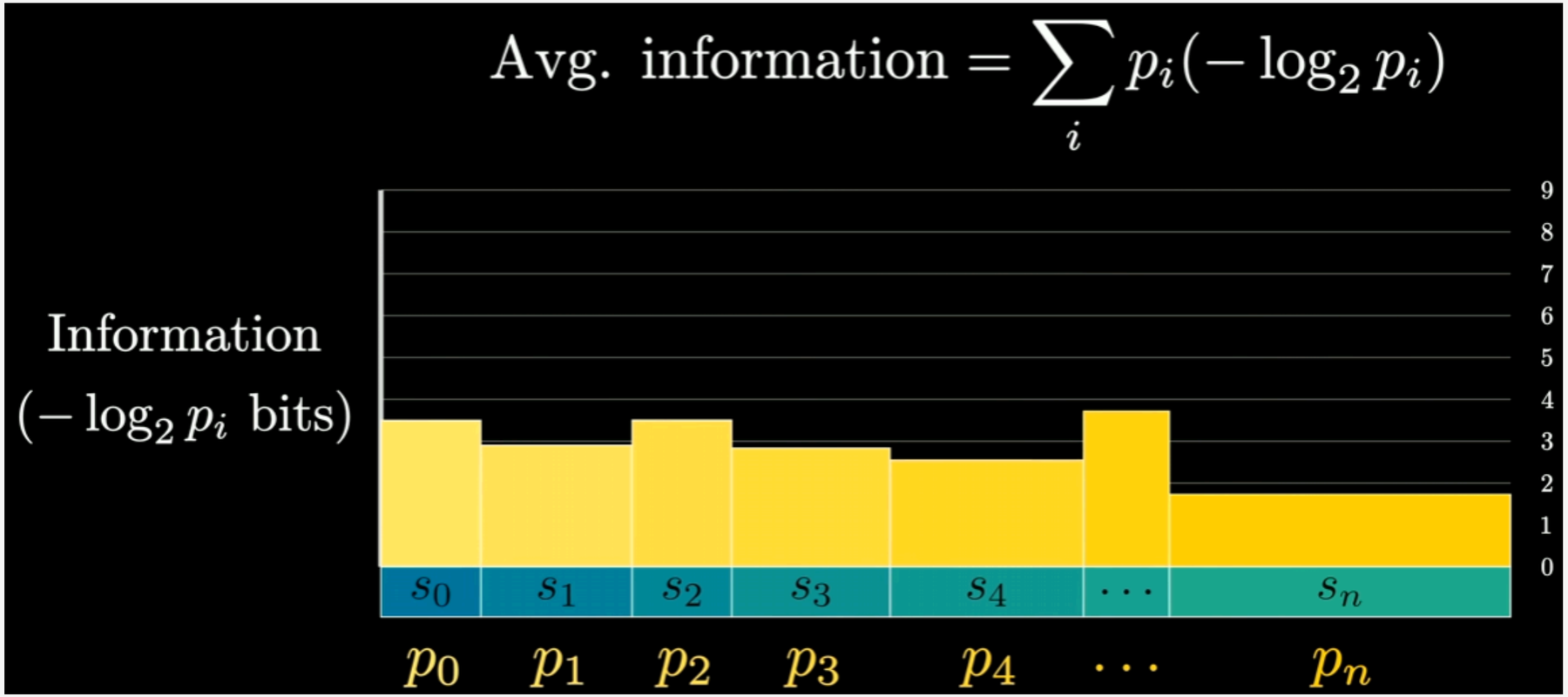

QISKIT
2026
2026
FALL FEST

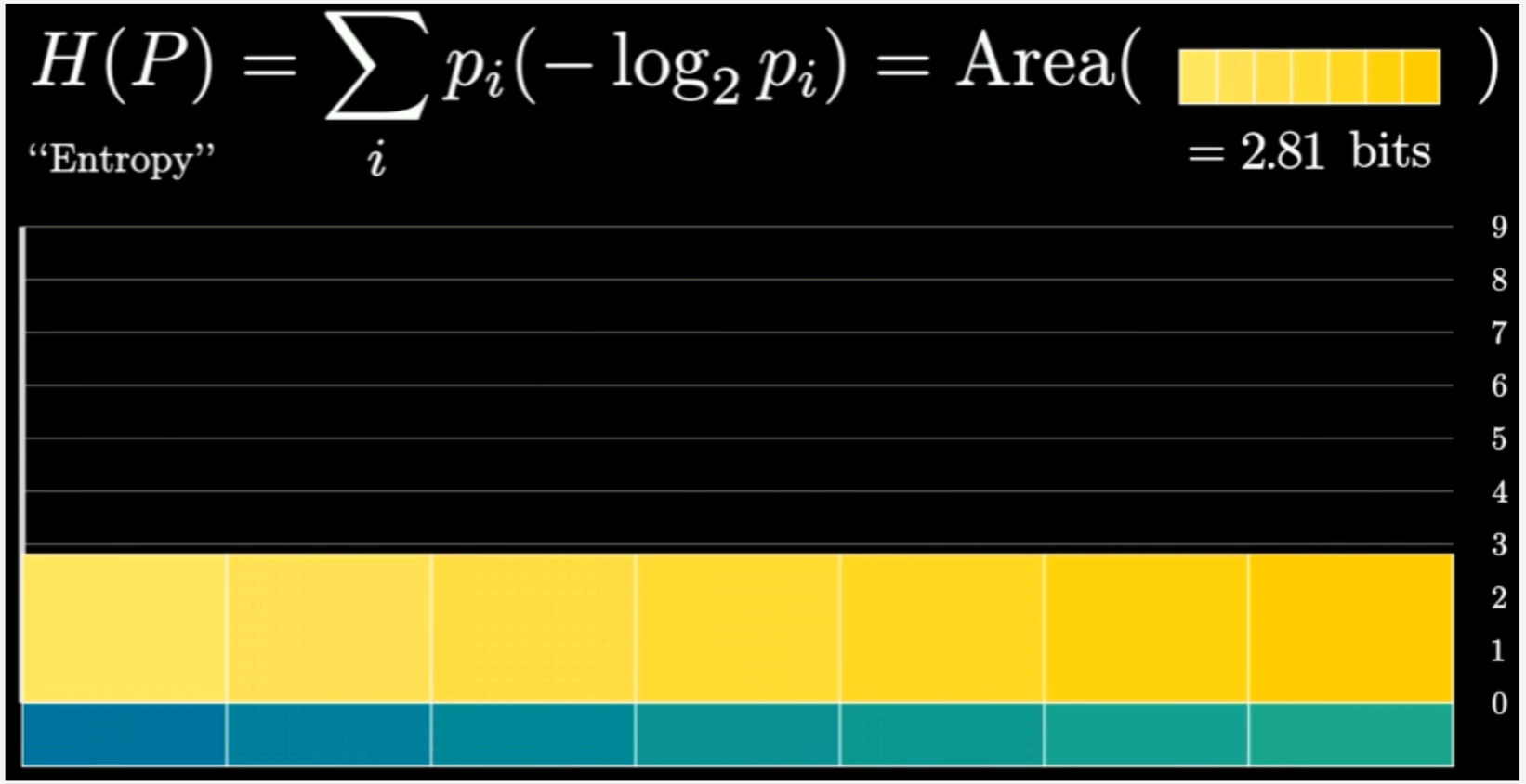

Random Noise is *incompressible*
11100110011000110101111111111110
01010110010100111100100100100111
00010100101000101000010001100111
00110111010110000100100100111000
01101000110111111101011010101100
10011000010011100101010100001110
01011001101100111011000001110100
00001001001110011101001000100010
10110011000111110101101100111001
01011100001100100110110100001100
10011011010110001111101000110110
10100010000000001011000000111111
11000000010011111100110111100010
00001111001111100100101100110000
00100101110001111001001110011001
01001101001111001100011010100101

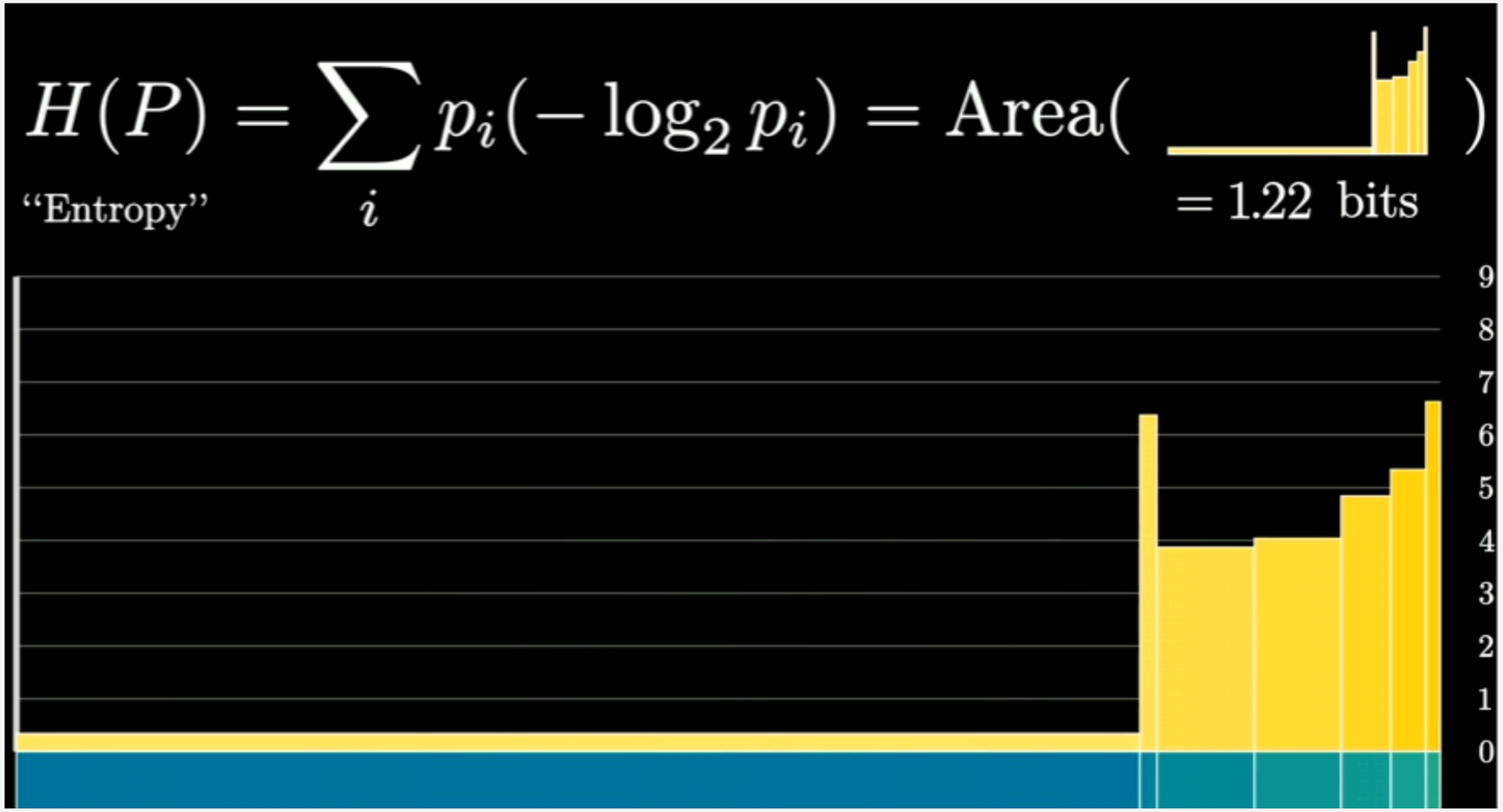

$\mathbf{P}(\text{any } n\text{-bit message}) = \frac{1}{2^n}$
$-\log_2(P(\text{“INFORMATION THEORY”}))$
$-\log_2(p) = n$
$-\log_2(p_0) - \log_2(p_1) - \log_2(p_2) - \cdots - \log_2(p_n)$

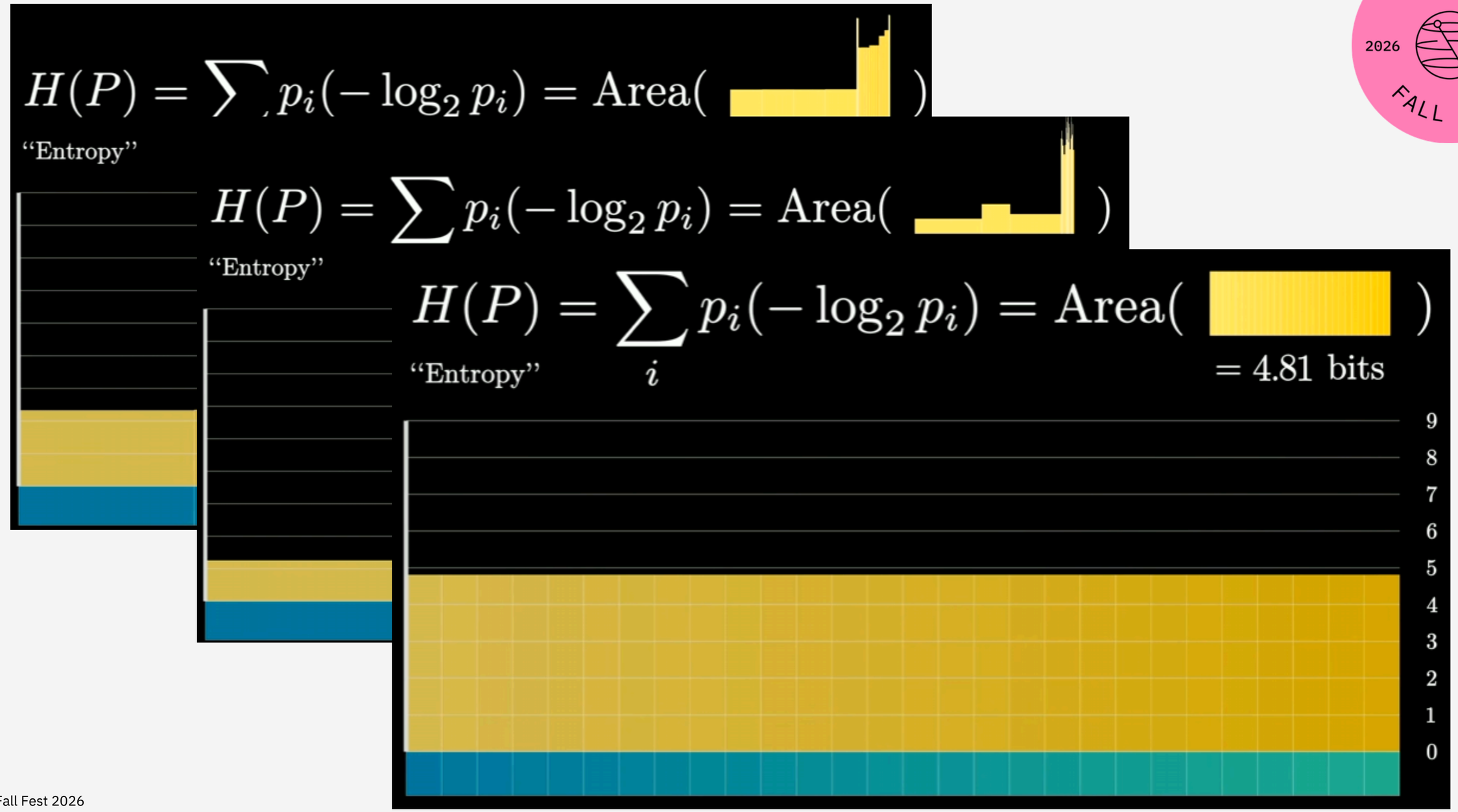

10101010000010100101000000001

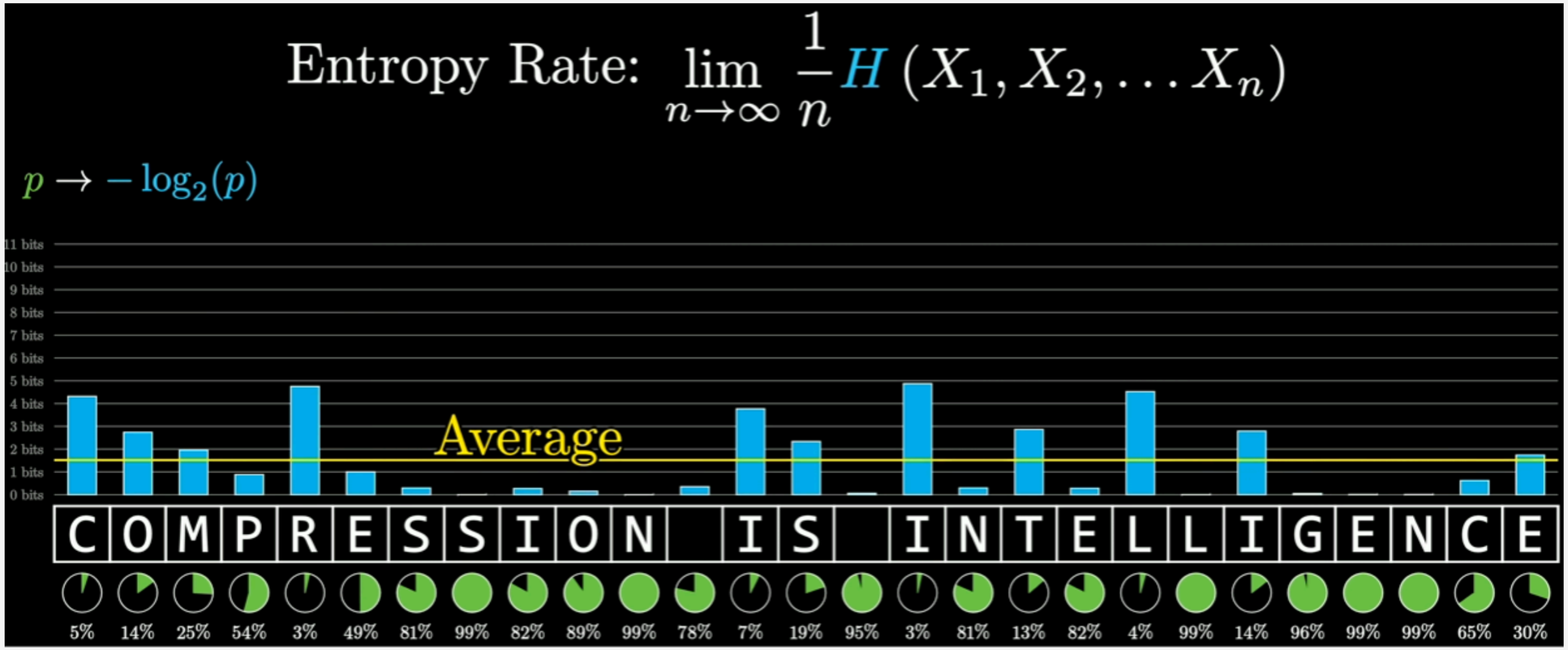

$\frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 + \frac{1}{8} \cdot 3 = 1.75$ bits
Information
$(-\log_2 p_i$ bits)
1 bit
2 bits
3 bits
3 bits
0
10
110
111
9
8
7
6
5
4
3
2
1
0
$\frac{1}{2}$
$\frac{1}{4}$
$\frac{1}{8}$
$\frac{1}{8}$


Avg. information $= \sum_i p_i(-\log_2 p_i)$
Information
$(-\log_2 p_i$ bits)
9
8
7
6
5
4
3
2
1
0
s_0 s_1 s_2 s_3 s_4 $\cdots$ s_n
p_0 p_1 p_2 p_3 p_4 $\cdots$ p_n


$H(P) = \sum_i p_i(-\log_2 p_i) = \text{Area}($)
"Entropy"
$= 2.81$ bits
9
8
7
6
5
4
3
2
1
0

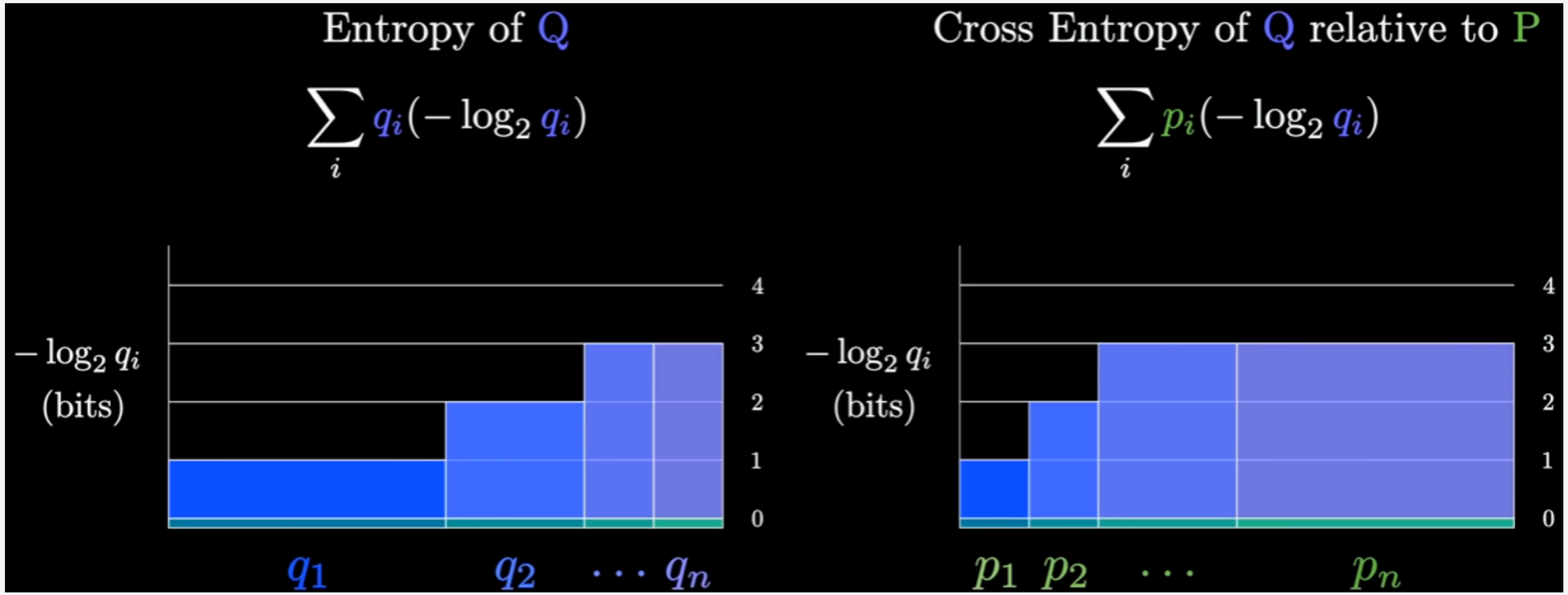

$H(P) = \sum_i p_i(-\log_2 p_i) = \text{Area}($)
"Entropy"
$= 1.22$ bits
9
8
7
6
5
4
3
2
1
0

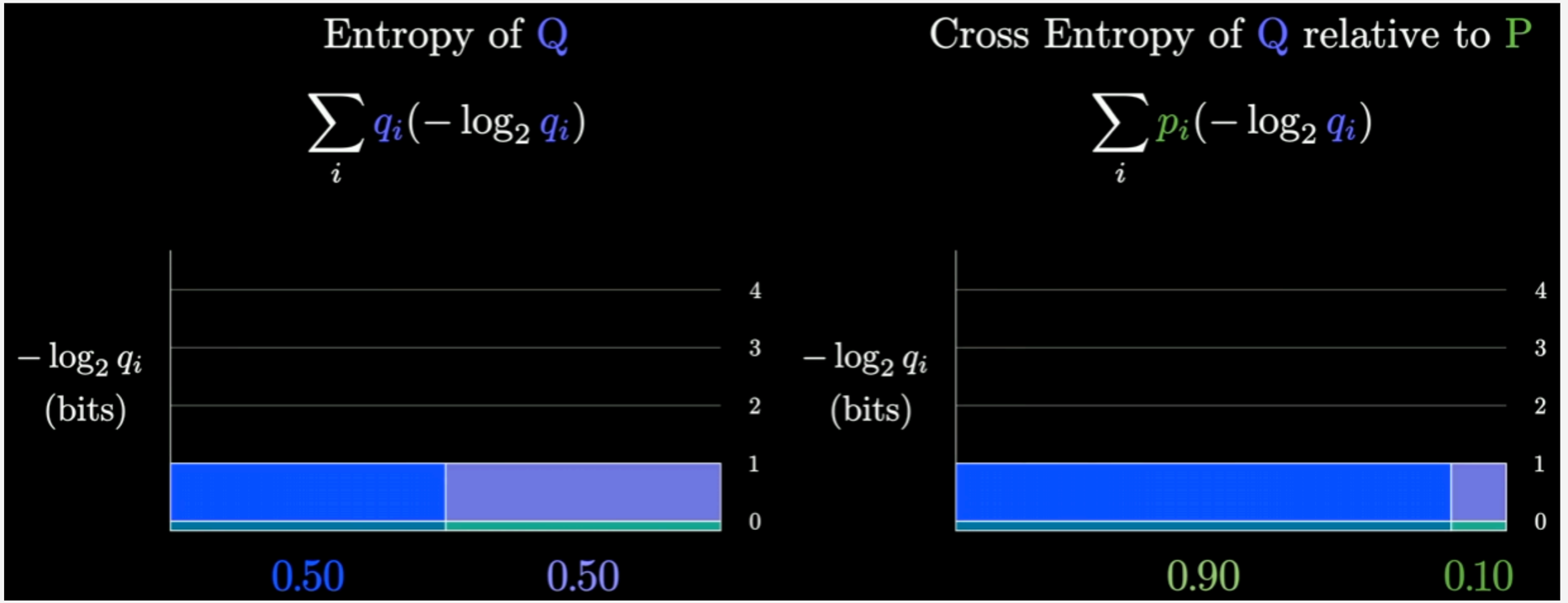

$H(P) = \sum_i p_i(-\log_2 p_i) = \text{Area}($)
"Entropy"
$H(P) = \sum_i p_i(-\log_2 p_i) = \text{Area}($)
"Entropy"
$H(P) = \sum_i p_i(-\log_2 p_i) = \text{Area}($)
"Entropy"
$= 4.81$ bits
9
8
7
6
5
4
3
2
1
0

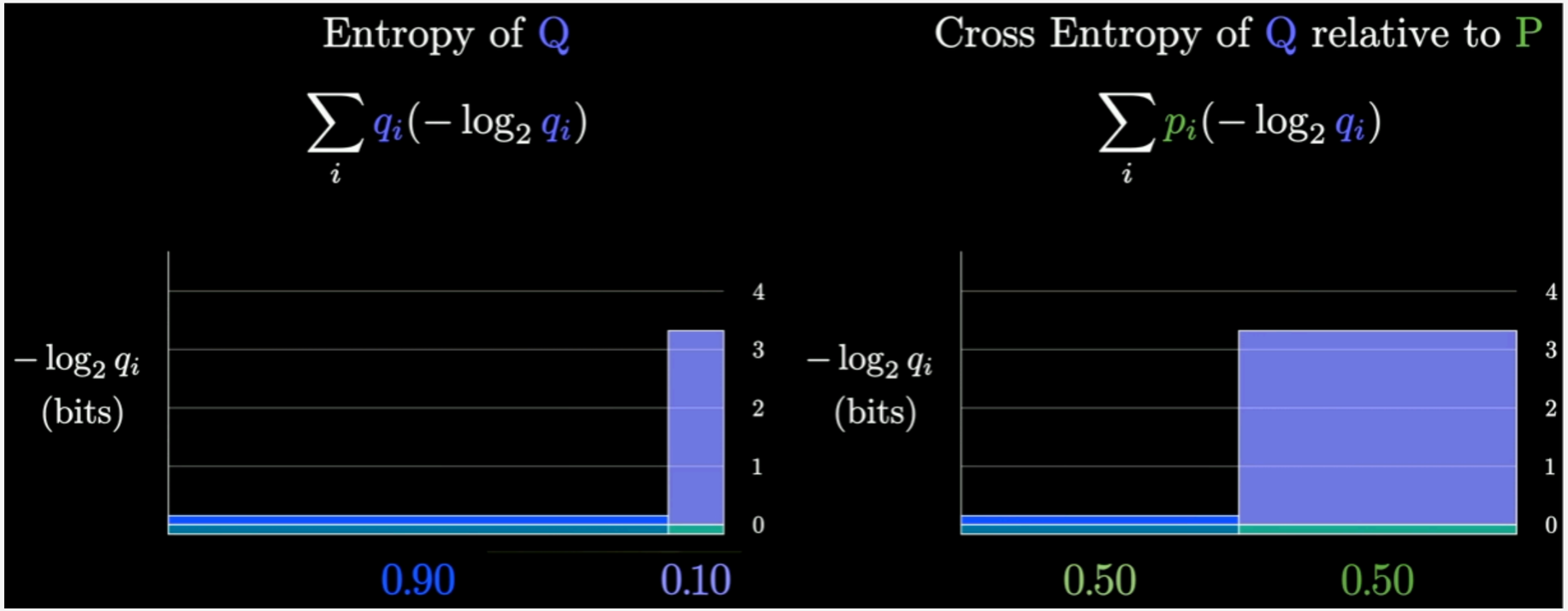

Entropy Rate: $\lim_{n\to\infty} \frac{1}{n} H(X_1, X_2, \ldots X_n)$
$p \to -\log_2(p)$
11 bits
10 bits
9 bits
8 bits
7 bits
6 bits
5 bits
4 bits
3 bits
2 bits
1 bits
0 bits
Average
COMPRESSION IS INTELLIGENCE
5% 14% 25% 54% 3% 49% 81% 99% 82% 89% 99% 78% 7% 19% 95% 3% 81% 13% 82% 4% 99% 14% 96% 99% 99% 65% 30%

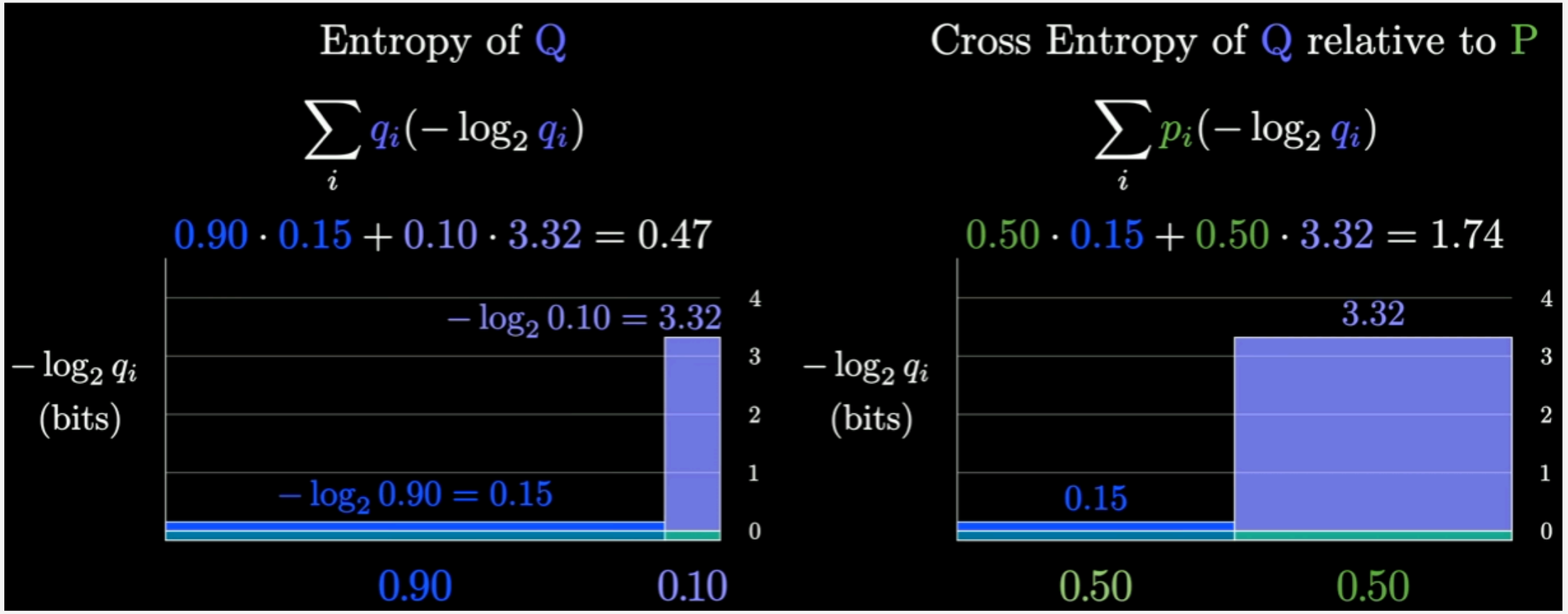


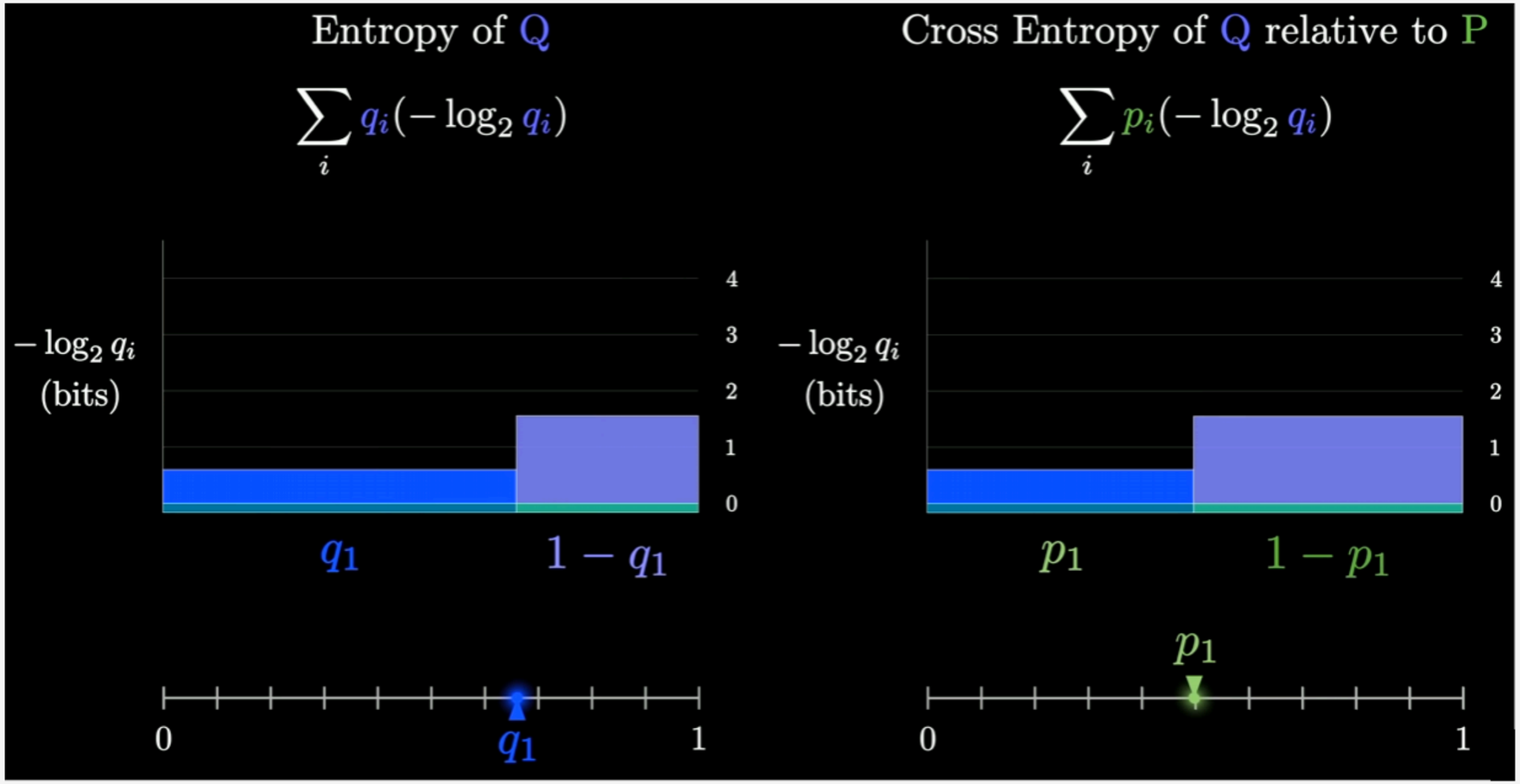
BACK TO THE BOARD

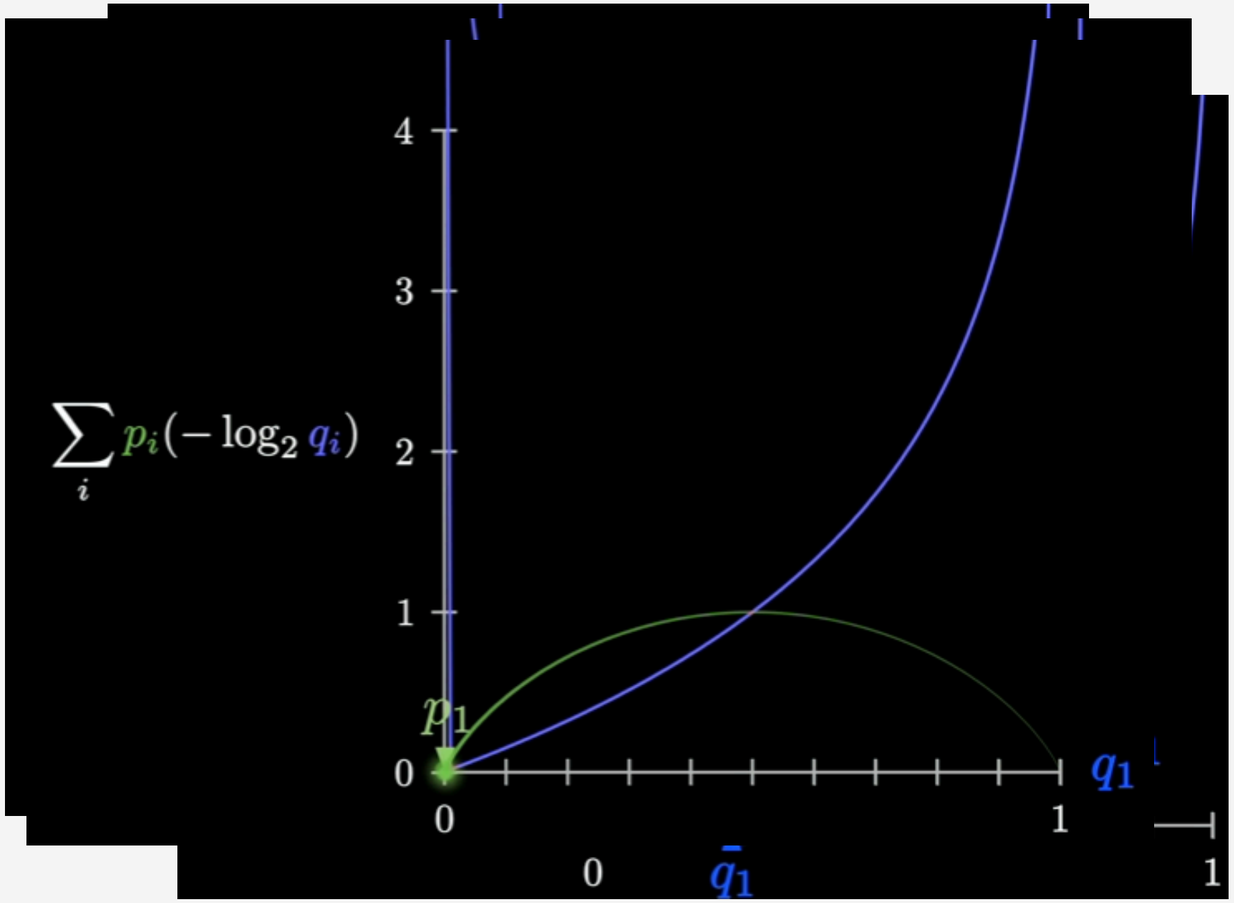

Entropy of Q
$\sum_i q_i(-\log_2 q_i)$
$-\log_2 q_i$
(bits)
4
3
2
1
0
q_1
q_2
$\cdots$
q_n
Cross Entropy of Q relative to P
$\sum_i p_i(-\log_2 q_i)$
$-\log_2 q_i$
(bits)
4
3
2
1
0
p_1
p_2
$\cdots$
p_n

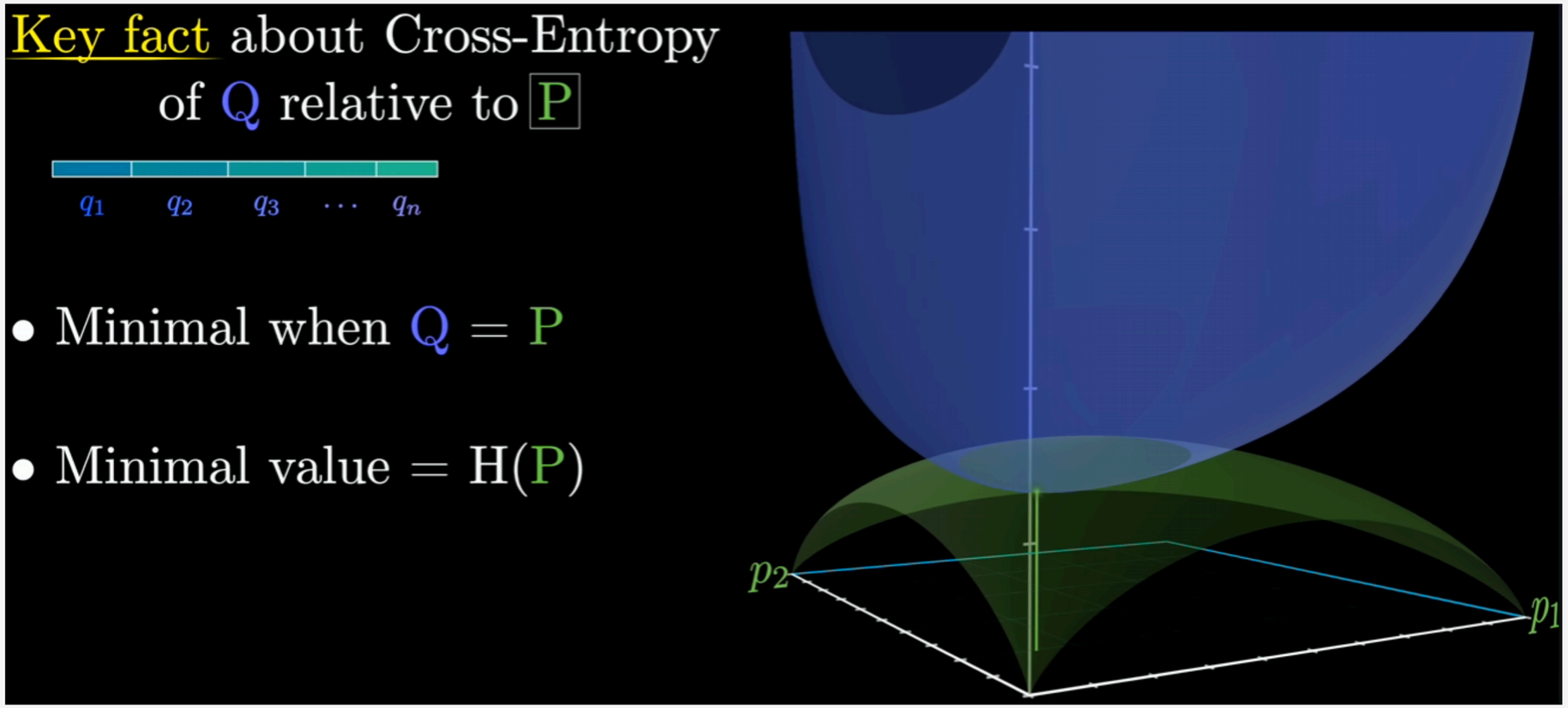

Entropy of Q
$\sum_i q_i(-\log_2 q_i)$
$-\log_2 q_i$
(bits)
4
3
2
1
0
0.50
0.50
Cross Entropy of Q relative to P
$\sum_i p_i(-\log_2 q_i)$
$-\log_2 q_i$
(bits)
4
3
2
1
0
0.90
0.10

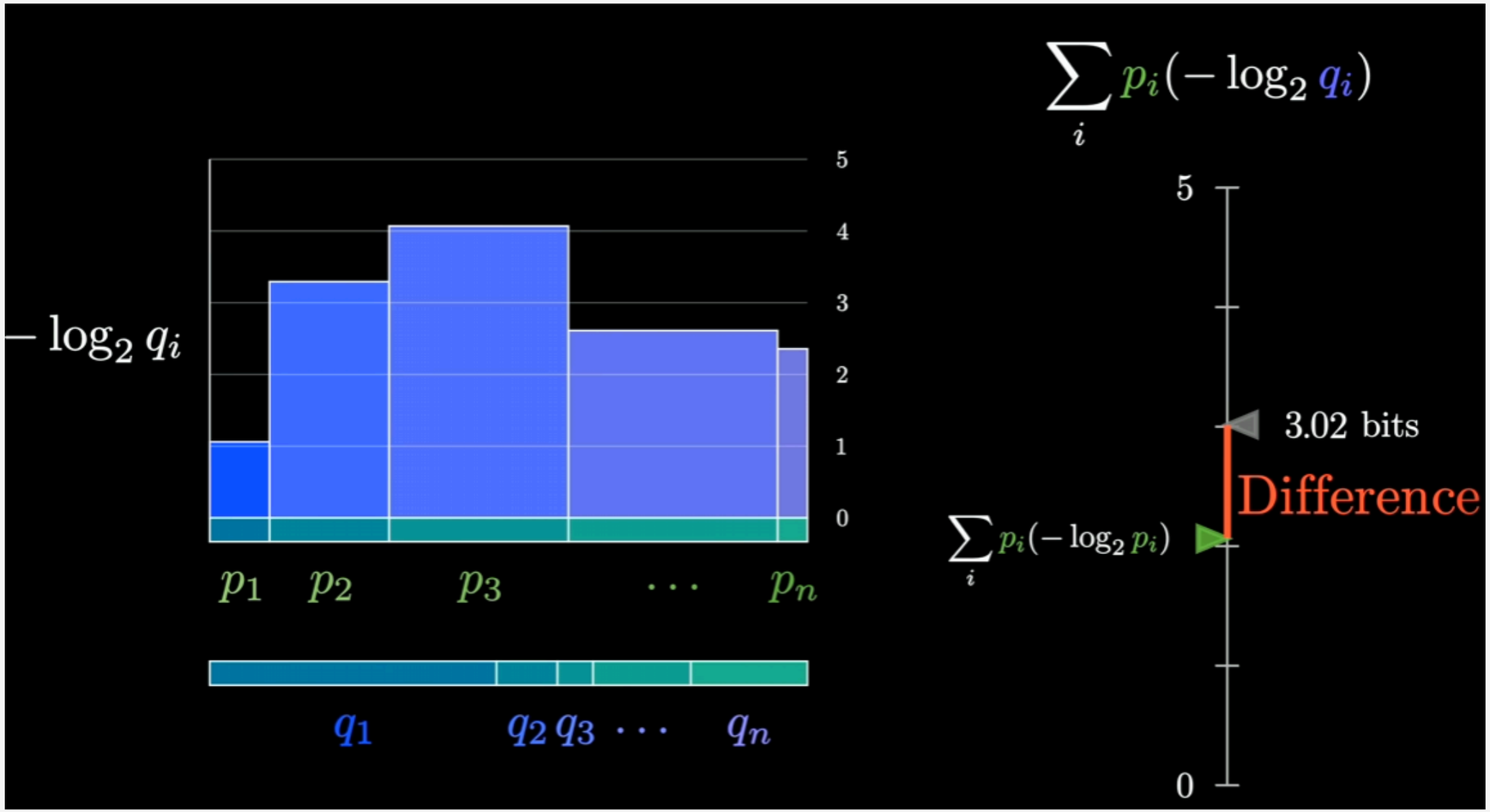

Entropy of Q
$\sum_i q_i(-\log_2 q_i)$
$-\log_2 q_i$
(bits)
4
3
2
1
0
0.90
0.10
Cross Entropy of Q relative to P
$\sum_i p_i(-\log_2 q_i)$
$-\log_2 q_i$
(bits)
4
3
2
1
0
0.50
0.50


Entropy of Q
$\sum_i q_i(-\log_2 q_i)$
$0.90 \cdot 0.15 + 0.10 \cdot 3.32 = 0.47$
$-\log_2 q_i$ (bits)
$-\log_2 0.10 = 3.32$
$-\log_2 0.90 = 0.15$
0.90
0.10
Cross Entropy of Q relative to P
$\sum_i p_i(-\log_2 q_i)$
$0.50 \cdot 0.15 + 0.50 \cdot 3.32 = 1.74$
$-\log_2 q_i$ (bits)
3.32
0.15
0.50
0.50
0
1
2
3
4


Entropy of Q
$\sum_i q_i(-\log_2 q_i)$
$-\log_2 q_i$
(bits)
4
3
2
1
0
q_1
$1-q_1$
0
q_1
1
Cross Entropy of Q relative to P
$\sum_i p_i(-\log_2 q_i)$
$-\log_2 q_i$
(bits)
4
3
2
1
0
p_1
$1-p_1$
p_1
0
1


$\sum_i p_i(-\log_2 q_i)$
4
3
2
1
0
p_1
0
1
q_1
0
$\bar{q}_1$
1


Key fact about Cross-Entropy
of Q relative to P
q_1 q_2 q_3 $\cdots$ q_n
• Minimal when Q = P
• Minimal value = H(P)
p_2
p_1

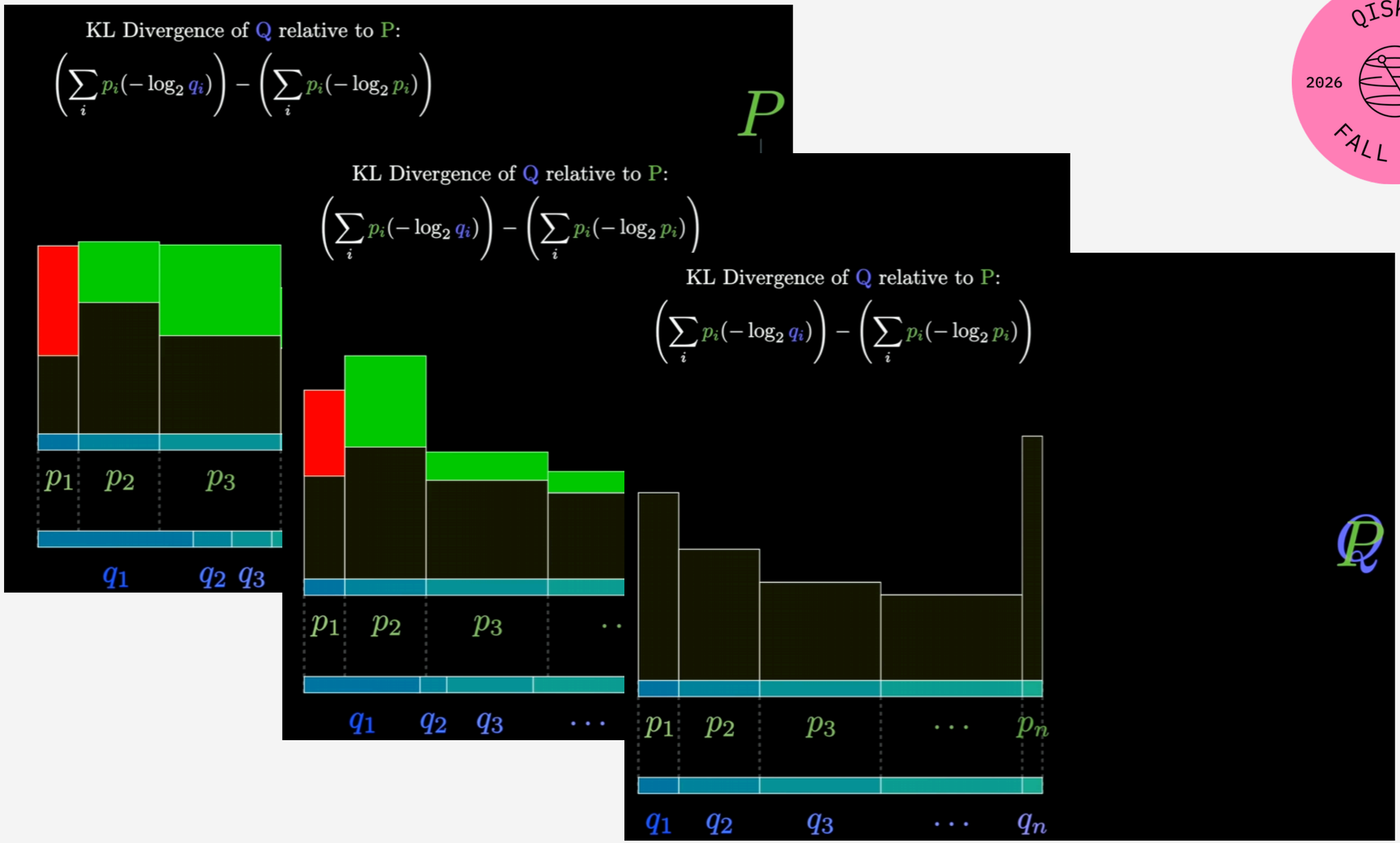

$\sum_i p_i(-\log_2 q_i)$
$-\log_2 q_i$
5
4
3
2
1
0
p_1 p_2 p_3 $\cdots$ p_n
q_1 q_2 q_3 $\cdots$ q_n
5
3.02 bits
Difference
$\sum_i p_i(-\log_2 p_i)$
0

Kullback-Leibler Divergence:

$$\left(\sum_i p_i(-\log_2 q_i)\right) - \left(\sum_i p_i(-\log_2 p_i)\right)$$

Cross Entropy of Q relative to P

Entropy of P


KL Divergence of Q relative to P:
$\left(\sum_i p_i(-\log_2 q_i)\right) - \left(\sum_i p_i(-\log_2 p_i)\right)$
P
KL Divergence of Q relative to P:
$\left(\sum_i p_i(-\log_2 q_i)\right) - \left(\sum_i p_i(-\log_2 p_i)\right)$
p_1 p_2 p_3
q_1 q_2 q_3
KL Divergence of Q relative to P:
$\left(\sum_i p_i(-\log_2 q_i)\right) - \left(\sum_i p_i(-\log_2 p_i)\right)$
p_1 p_2 p_3 $\cdots$
q_1 q_2 q_3 $\cdots$
p_1 p_2 p_3 $\cdots$ p_n
q_1 q_2 q_3 $\cdots$ q_n
Q


KL Divergence of Q relative to P:
$\left(\sum_i p_i(-\log_2 q_i)\right) - \left(\sum_i p_i(-\log_2 p_i)\right)$
$\neq$
KL Divergence of P relative to Q:
$\left(\sum_i q_i(-\log_2 p_i)\right) - \left(\sum_i q_i(-\log_2 q_i)\right)$

Part II
Quantum Information Theory



Quantum information theorems

1. No-teleportation theorem, which states that a qubit cannot be (wholly) converted into classical bits; that is, it cannot be fully "read".

2. No-cloning theorem, which prevents an arbitrary qubit from being copied.

3. No-deleting theorem, which prevents an arbitrary qubit from being deleted.

4. No-broadcast theorem, which prevents an arbitrary qubit from being delivered to multiple recipients, although it can be transported from place to place (e.g. via quantum teleportation).

5. No-hiding theorem, which demonstrates the conservation of quantum information.

Von Neumann Entropy

$$S(\hat{\rho}) = -\langle \log \hat{\rho} \rangle = -\mathrm{Tr}(\hat{\rho} \log(\hat{\rho}))$$

Properties of Von Neumann Entropy

- Zero
- Maximal
- Invariant under Unitary
- Concave
- Additive for indpendent systems
- Strongly subadditive

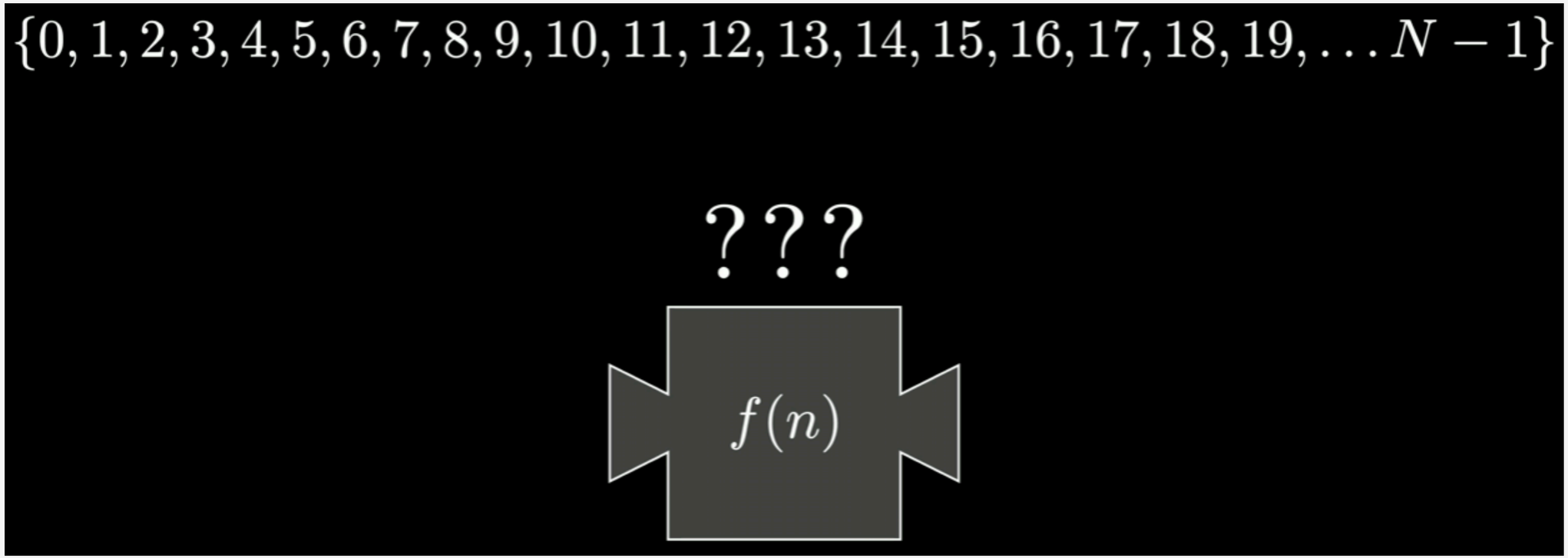
Quantum Cross Entropy and Quantum Relative Entropy (Umegaki Relative Entropy)

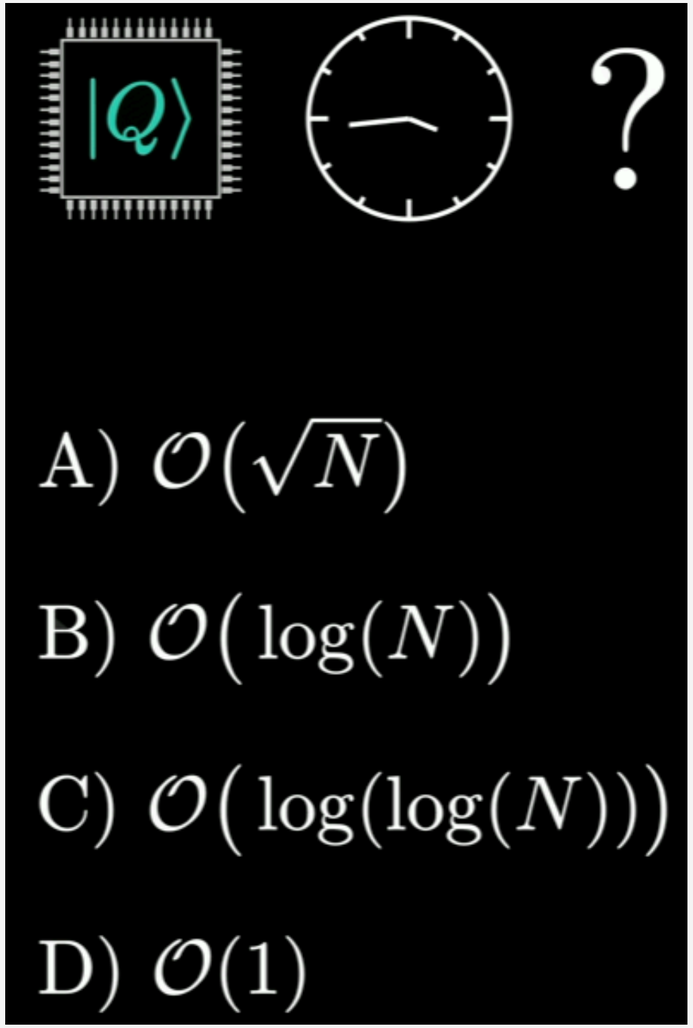
$$S(\rho,\sigma)=-\mathbf{Tr}(\boldsymbol{\rho}\log\boldsymbol{\sigma})$$

$$D(\rho /\!\!/ \sigma)=\mathrm{Tr}(\boldsymbol{\rho}(\log\boldsymbol{\rho}-\log\boldsymbol{\sigma}))$$

Part III
What exactly is ‘Quantum Computing’?




$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, \ldots N-1\}$
???
$f(n)$


$|Q\rangle$
?
A) $\mathcal{O}(\sqrt{N})$
B) $\mathcal{O}(\log(N))$
C) $\mathcal{O}(\log(\log(N)))$
D) $\mathcal{O}(1)$



BOARD PLEASE!

Part IV
Where can we apply Quantum Computing?



Quantum Computers are NOT omnipotent

So where can we apply quantum computers?

Active research going on, but prominent fields:

- Optimization
- Quantum sensing and metrology
- Quantum networks and communications

Closing Remarks: Quantum Computers and privacy



Thank You!